

# Condominium Management System

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# Vanguard(Condominium Management System)

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## Abstract

The Vanguard Condominium Management System was developed to digitalize and streamline the daily operations of residential communities by replacing manual, paper-based workflows with a unified, secure, and cloud-driven platform. The project addresses long-standing challenges such as inaccurate visitor tracking, fragmented communication, booking conflicts, manual document verification, and the administrative burden faced by property management teams. Vanguard integrates a mobile application built using Flutter and Firebase, providing residents, tenants, administrators, and security personnel with seamless access to essential services. The system comprises six major modules: Resident & Tenant Management, Visitor & Security Access, Facility Booking, Resident Services, Admin Management, and Payment Management, offering comprehensive functionality for identity verification through OCR and facial recognition, tenant lifecycle management, QR-based visitor entry, real-time facility scheduling, announcement broadcasting, complaint handling, and digital payment tracking. The development applied the Waterfall methodology, supported by UML-based design, Firebase backend services, Google ML Kit for AI processing, and structured architectural modelling. Extensive testing—including functional validation, usability checks, performance assessment, and AI accuracy evaluation—demonstrated the system's ability to deliver reliable document verification, efficient booking workflows, secure access control, and responsive communication.

The Vanguard system significantly enhances transparency, operational efficiency, and security within condominium environments while reducing administrative workload and improving resident experience. Although certain limitations exist, such as dependency on cloud infrastructure and variations in AI accuracy under challenging conditions, the project showcases the practical application of modern mobile technologies and cloud services in community management. Future enhancements may include integrating IoT access devices, predictive maintenance analytics, and expanded automation to further support smart condominium ecosystems.

## **Acknowledgement**

This contains acknowledgement to those who have contributed directly or indirectly to the completion of the project. Usually the people to be acknowledged include the project supervisor(s), moderators, family, and those who have given assistance and support to ensure the success of the project.

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# Chapter 1

## **Introduction**

# 1 Introduction

This chapter provides a comprehensive overview of the Condo Management System called “Vanguard” project. It outlines the background and purpose of the project, clearly defines the problem statement it aims to address, and establishes the specific research objectives and questions that will guide the development process. A critical component of this chapter is the delineation of the project scope, which explicitly states the functionalities that will be implemented within the confines of this project and those that will be excluded or handled by complementary modules. Furthermore, the chapter presents the project milestones and schedule, lists the software and hardware required for development, and concludes with a summary of the key points discussed.

This project is a collaborative effort between two developers. The primary author of this document is solely responsible for the design, implementation, and documentation of the following three core modules:

1. **Resident & Tenant Management Module**
2. **Visitor & Security Access Module**
3. **Facility Booking Module**

The remaining modules – namely the **Resident Services Module, Admin Management Module, and Payment Module** – are being developed by a teammate. While these modules are integral to the complete system and will be mentioned in the overall architectural context to provide a holistic view, the primary focus of this report, from requirements analysis to testing, will be on the three modules listed above. The scope of work for this individual project is strictly confined to these modules.

## 1.1 Project Background and Purpose

The management of modern condominiums and gated communities involves complex coordination between residents, property managers, and security personnel. Traditional, paper-based systems and disparate digital tools have proven inadequate, leading to operational inefficiencies, security vulnerabilities, and resident dissatisfaction. The purpose of this project, dubbed "Vanguard", is to develop a unified, mobile-first management system that seamlessly integrates six critical operational modules onto a single platform. By leveraging modern cross-platform development frameworks and cloud technologies, this project aims to digitize and automate core property management workflows. The overarching purpose is to create a scalable, secure, and user-centric solution that significantly reduces administrative overhead for management teams while enhancing convenience and safety for residents. This project serves as a practical implementation of software engineering principles to solve

a real-world problem, demonstrating the viability of integrated mobile solutions in the proptech (property technology) domain.

### **Support for Sustainable Development Goals (SDGs)**

This project aligns with the **United Nations Sustainable Development Goals (SDGs)**, particularly **SDG 11: Sustainable Cities and Communities** and **SDG 10: Reduced Inequalities**.

#### **For SDG 11 (Sustainable Cities and Communities):**

The system promotes efficient, transparent, and inclusive condominium management by digitalizing daily operations such as visitor registration, facility booking, and maintenance coordination. By integrating AI verification, QR access, and mobile-based interactions, the project enhances community safety, optimizes resource usage, and supports sustainable urban living. These features collectively contribute to building smart, secure, and well-managed residential communities that align with modern city development initiatives.

#### **For SDG 10 (Reduced Inequalities):**

The system ensures inclusivity by integrating accessibility features for **elderly and disabled residents**, allowing equal participation in community activities. Through a user-friendly mobile interface, everyone—regardless of age or ability—can access booking services, make payments, or submit maintenance requests. This helps **bridge the digital divide** within urban populations and promotes fairness, transparency, and equal access to digital services in modern housing environments.

### **Scope of the Project**

The system consists of six key modules:

1. **Resident & Tenant Management Module** – handles resident registration, tenant verification, and account lifecycle management.
2. **Visitor & Security Access Module** – manages QR-based entry passes, facial recognition, and visitor blacklisting.
3. **Facility Booking Module** – supports real-time scheduling, automated approvals, and notifications.

## **1.2 Problem Statement**

The existing paradigm of condominium management is predominantly reliant on legacy systems characterized by manual processes and disconnected tools. This operational model presents significant, interlinked challenges that compromise efficiency, security, and resident satisfaction. The core problems identified are as follows:

### **1.2.1 Inefficient and Insecure Visitor Tracking**

The industry-standard practice of using paper-based log books for visitor management is fundamentally flawed. These physical logs are susceptible to a range of issues including poor handwriting, incomplete entry fields, and intentional data omission or forgery, which severely compromises the integrity and reliability of visitor data (Xenia Team, 2024). This lack of reliable data creates critical security vulnerabilities, as security personnel cannot quickly or accurately verify the identity of individuals seeking entry, increasing the risk of unauthorized access. Furthermore, the static nature of paper logs provides no capability for real-time monitoring or instant alerts, leaving security teams with outdated information.

### **1.2.2 Disorganized and Manual Facility Booking Process**

The prevalent use of phone calls, emails, and physical sign-up sheets for reserving common facilities such as gyms, pools, and function rooms is highly inefficient. This decentralized and informal approach inherently leads to scheduling conflicts and double-bookings, as there is no single source of truth for availability (Harish Kannan R, 2025). These conflicts result in resident disputes and dissatisfaction. Moreover, this process places a substantial administrative burden on property management staff, who are required to manually coordinate, record, and reconcile these bookings, consuming time that could be allocated to higher-value tasks (Bookingplanner, 2022).

### **1.2.3 Fragmented Communication and Maintenance Reporting**

A critical gap exists in the communication channel between residents and property management. Residents often lack a direct, dedicated, and reliable method to report maintenance issues or provide feedback. Consequently, requests are frequently lost within crowded email inboxes or forgotten phone messages, leading to significant delays in response and resolution times (Timetec, 2025a). This fragmentation fosters a sense of neglect among residents and hinders the management's ability to address problems proactively and maintain the property effectively.

### **1.2.4 Lack of Proactive Security Measures and Centralized Oversight**

Current systems fail to provide a mechanism for proactive security enforcement. There is no centralized or digital means to blacklist suspicious visitors or track recurring offenders, allowing

potential security threats to persist unchecked (Timetec, 2025b). Additionally, the reliance on traditional keycards or PIN codes for resident access is outdated; these credentials can be easily shared, lost, or stolen, further undermining the security of the premises. The absence of a unified digital platform also means there is no holistic view of operations for administrators, preventing data-driven decision-making.

These interconnected problems collectively illustrate an urgent need for a consolidated, automated, and mobile-centric solution that can seamlessly integrate these core management functions into a single, secure, and user-friendly platform.

### **1.3 Research Objectives**

The overarching aim of this project is to design, develop, and evaluate a comprehensive mobile-based condominium management system named "Vanguard" that addresses the critical problems identified in Section 1.2. To achieve this aim, the following specific research objectives have been formulated:

#### **1.3.1 To design and develop a secure and automated access control system.**

This objective directly addresses problem 1.2.1 (Inefficient and Insecure Visitor Tracking). The focus will be on implementing a dual-mode access system utilizing QR-code-based JWT tokens for visitors and on-device facial recognition for residents (using ML Kit). The outcome will be a subsystem that enhances security by providing verifiable, time-bound credentials and touchless entry, while maintaining a real-time digital audit log of all entry and exit events.

#### **1.3.2 To implement an intelligent and conflict-free facility booking module.**

This objective is formulated to solve problem 1.2.2 (Disorganized and Manual Facility Booking Process). The module will feature a real-time scheduling system with automated conflict detection algorithms to prevent double-bookings. It will incorporate an automated approval workflow to streamline management tasks. The successful implementation of this objective will result in optimized facility utilization, reduced administrative overhead, and an improved resident experience.

#### **1.3.3 To create a centralized digital platform for resident identity management and communication.**

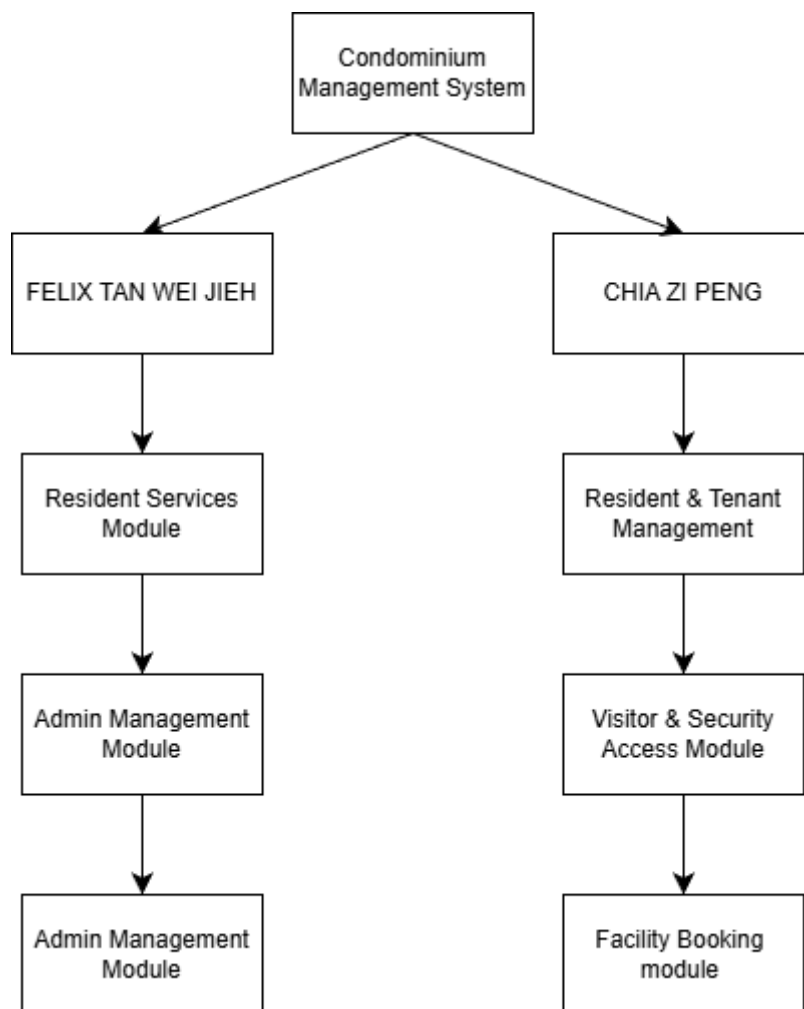
This objective tackles problem 1.2.3 (Fragmented Communication) and part of 1.2.4 (Lack of Oversight). The core is the development of a KYC (Know Your Customer) verification process using OCR (Optical Character Recognition) technology to automate and secure the onboarding of owners and tenants by validating property deeds and identification documents. Furthermore, this objective includes integrating real-time in-app chat functionality to bridge the communication gap between residents, administrators, and security personnel, ensuring timely and logged interactions.

## 1.4 Research Questions

1. How does AI and facial recognition improve security and prevent unauthorized access in condominium management?
2. How can automation in booking, maintenance, and payment help reduce admin work and improve efficiency?
3. How does a mobile-based management system affect resident satisfaction, communication, and user experience compared to traditional methods?

## 1.5 Project Research Scope

This section defines the boundaries of the "Vanguard" project by detailing the functionalities included within the scope of development. The project is a collaborative effort, and the responsibilities are divided between me (Chia Zi Peng) and teammate (Felix Tan Wei Jieh) as outlined in the table below.



**Figure 1.1 : Overview of Module Breakdown**

Figure 1.1: Overview of Project Module Distribution illustrates the architectural structure of the SmartResident residential management platform, which is organized into six core functional modules. The development responsibilities are divided between two team members: Me, Chia Zi Peng, responsible for three modules, while my teammate, Felix Tan Wei Jieh, is responsible for the remaining three. The diagram further clarifies the user-facing platforms, distinguishing features designed for the resident mobile application from those built for the administrative web and mobile dashboard, and shows their integration via a centralized Firebase backend.

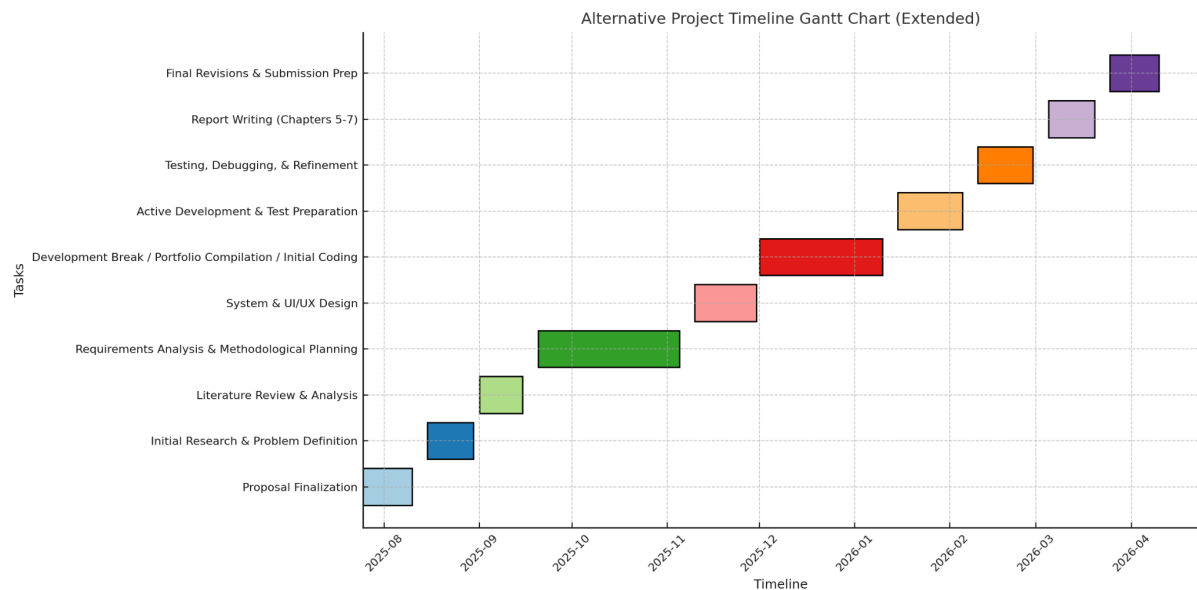
| Name                | Module                          | In-Scope Features & Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------|---------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Chia Zi Peng</b> | 1. Resident & Tenant Management | <ul style="list-style-type: none"><li>• Digital Onboarding: A system for owners to register by uploading proof of ownership (e.g., title deeds).</li><li>• Automated KYC Verification: Integration of OCR (Optical Character Recognition) technology to automatically validate uploaded documents for authenticity.</li><li>• Tenant Account Lifecycle: Functionality for owners to create and manage tenant accounts linked to lease periods, with auto-expiry.</li><li>• Profile Management: Full CRUD (Create, Read, Update, Delete) operations for user profiles, managed via role-based access control.</li></ul>                                                                                                                                                |
|                     | 2. Visitor & Security Access    | <ul style="list-style-type: none"><li>• Visitor Pre-registration: A process for residents to generate time-bound, cryptographically signed QR code passes for their guests within the app.</li><li>• Security Verification Interface: A dedicated screen for security guards to scan QR codes, verify visitor identity, and log all entry/exit events with timestamps to Firestore.</li><li>• Resident Facial Recognition: Implementation of a secure, on-device facial recognition system (using Firebase ML Kit) for touchless resident entry, including user enrollment and authentication flows.</li><li>• Blacklist Integration: Providing the interface and data structure for security to flag and prevent blacklisted visitors from gaining access.</li></ul> |

|                           |                             |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                           | 3. Facility Booking         | <ul style="list-style-type: none"> <li>• <b>Real-time Booking CRUD:</b> Residents can view availability, book, reschedule, and cancel facility reservations (e.g., gym, pool).</li> <li>• <b>Conflict Detection:</b> Implementation of algorithms to prevent double-booking and scheduling conflicts in real-time.</li> <li>• <b>Approval Workflows:</b> Automated notifications and a streamlined interface for administrators/security to review and approve or reject booking requests.</li> <li>• <b>User Notifications:</b> Integration of push notifications for booking status updates (confirmation, rejection, reminders).</li> </ul>                                                                                                                                                                                                                                                                                                                           |
| <b>Felix Tan Wei Jieh</b> | 1. Resident Services Module | <p><b>Key Features:</b></p> <ul style="list-style-type: none"> <li>• <b>Maintenance Request Hub:</b> Residents can submit detailed maintenance requests with photo/video evidence. Each request is tracked in real time with clear status updates (“Pending” → “In Progress” → “Resolved”). This reduces miscommunication and ensures accountability.</li> <li>• <b>Emergency Alert System:</b> A one-tap alert instantly notifies security personnel with the resident’s exact unit number and location. For accessibility, large buttons and call activation support (for elderly or disabled users) are included. Alerts are encrypted for resident privacy.</li> <li>• <b>Community Forum:</b> A secure, moderated discussion space for neighborhood announcements and resident discussions. Built with real-time Firestore updates, with moderation tools to prevent spam or misuse. The forum strengthens community bonds and encourages participation.</li> </ul> |
|                           | 2. Admin Management Module  | <p><b>Key Features:</b></p> <ul style="list-style-type: none"> <li>• <b>User Management:</b> Admins can approve/disable accounts, verify ownership documents using OCR (via Firebase ML Kit), and assign roles (Admin, Tenant, Owner,</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

|  |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|--|-------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  |                   | <p>Security). All sensitive documents are encrypted and stored securely to protect resident privacy.</p> <ul style="list-style-type: none"><li>● <b>Facility Scheduling:</b> A smart booking system with conflict detection prevents double bookings. Admins can dynamically edit availability hours, and residents receive instant confirmations via push notifications.</li><li>● <b>Data-Driven Insights:</b> Interactive dashboards display real-time analytics on maintenance backlogs, visitor traffic, and payment patterns. This allows proactive management and reduces operational risks.</li><li>● <b>Feedback Inbox:</b> Admins can receive, categorize, and respond to resident feedback efficiently, improving engagement and transparency.</li><li>● <b>Announcements:</b> Admins can post, update, or delete announcements, ensuring residents stay informed. Push notifications and in-app banners reduce the risk of missed communication.</li></ul> |
|  | 3. Payment Module | <ul style="list-style-type: none"><li>● <b>Technology Stack:</b> Ethereum Blockchain (Ganache test environment), Web3.js, Stripe/PayPal REST APIs, Firebase (secure transaction logging).</li><li>● <b>Core Functionality:</b> This module facilitates transparent and secure financial transactions. It integrates a blockchain-based payment ledger using Ethereum to ensure immutability and auditability for maintenance fees and other charges. A conventional payment gateway (Stripe/PayPal) is included as a fallback for reliability. Residents can access a verifiable payment history, while the system automates compliance tasks such as late-payment reminders and report generation, adhering to financial data protection standards.</li></ul>                                                                                                                                                                                                         |

**Table 1.1 Project Module Breakdown by Team Members**

## 1.6 Project Milestones and Schedule

**Figure 1.1 Project Gantt Chart**

| ACTIVITIES                                             | EXPECTED OUTCOME                                | COMPLETION DATE |
|--------------------------------------------------------|-------------------------------------------------|-----------------|
| Prepare and Submit Proposal (Form 1–3)                 | Official proposal submission                    | 08/08 /2025     |
| Submit Chapter 1 – Introduction                        | Background, objectives & scope of modules       | 22/08/2025      |
| Submit Chapter 2 – Research Background                 | Literature review & competitor analysis         | 08/09/2025      |
| Submit Chapter 3 – Methodology & Requirements Analysis | Detailed development plan regarding the project | 22/09/2025      |
| Submit Chapter 4 – System Design                       | Wireframes, mockups, architecture plan          | 21/11/2025      |
| Submit Project I Portfolio                             | Complete documentation (Chapter 1 - 4)          | 12/12/2025      |

|                                   |                                                |            |
|-----------------------------------|------------------------------------------------|------------|
| Prepare Test Plan / Preview       | Verified test scenario & user flows            | 20/02/2026 |
| Final System Testing              | Fully functional system                        | 06/03/2026 |
| Submit Draft FYP Report           | Chapters 1-7 with implementation details       | 03/04/2026 |
| Submit Final Report & Source Code | Complete FYP documentation and ready to submit | 13/04/2026 |

**Table 1.3 Project Schedule**

## 1.7 Software and Hardware Utilized

The development and testing of the "Vanguard" condominium management system will require the following software tools and hardware components.

### 1.7.1 Software

The following software technologies and services will be utilized for development, deployment, and project management:

| Category             | Tool / Technology      | Purpose                                                                                                 |
|----------------------|------------------------|---------------------------------------------------------------------------------------------------------|
| Frontend Development | Flutter SDK            | Primary framework for building the cross-platform mobile application for both residents and management. |
|                      | Dart                   | The programming language used with Flutter.                                                             |
| Backend & Database   | Firebase Auth          | To handle user authentication (resident, security, admin) and role-based access control.                |
|                      | Cloud Firestore        | NoSQL database for storing real-time data (user profiles, bookings, access logs, maintenance requests). |
|                      | Firebase Cloud Storage | To store user-uploaded documents (e.g., title                                                           |

|                   |                                |                                                                                               |
|-------------------|--------------------------------|-----------------------------------------------------------------------------------------------|
|                   |                                | deeds) and other media files.                                                                 |
|                   | Firebase Cloud Functions       | For running backend code in response to events (e.g., sending notifications, processing OCR). |
|                   | Firebase Cloud Messaging (FCM) | To send push notifications for booking updates, security alerts, etc.                         |
| AI/ML Services    | Firebase ML Kit                | For on-device facial recognition capabilities for resident access.                            |
|                   | Google Cloud Vision API        | For cloud-based OCR to verify uploaded identity and property documents.                       |
| Development Tools | Android Studio                 | Integrated Development Environments (IDEs) for writing and debugging code.                    |
|                   | GitHub                         | For version control and collaborative code management with the teammate.                      |

**Table 1.3 Software Utilized**

### 1.7.2 Hardware

**Laptop/Desktop** - The main development device used for coding, compiling, running emulators, and integrating all system modules.

**Smartphones (Android)** - Essential for real-device testing of the mobile app, ensuring usability, performance, and compatibility with different devices.

**Stable internet connection** - Required for real-time synchronization with Firebase, blockchain testing with Ganache, collaborative development on GitHub, and deployment processes.

## 1.8 Chapter Summary and Evaluation

This chapter has successfully laid a comprehensive and structured foundation for the "Vanguard" condominium management system project. It began by introducing the project's context and the collaborative nature of the development effort, clearly delineating the author's responsibility for the three core modules: Resident & Tenant Management, Visitor & Security Access, and Facility Booking.

The Project Background and Purpose effectively established the real-world necessity for this digital solution, highlighting the inefficiencies of traditional management methods and positioning "Vanguard" as a modern, integrated alternative aimed at enhancing both operational efficiency and resident satisfaction.

The Problem Statement was robustly defined, breaking down the core issues into four distinct and well-researched areas: inefficient visitor tracking, disorganized facility booking, fragmented communication, and a lack of proactive security. Each problem was substantiated with citations, demonstrating academic rigor and a clear understanding of the domain's pain points.

The Research Objectives were precisely formulated to directly address the stated problems, creating a clear line of sight from issue to solution. The objectives are Specific, Measurable, Achievable, Relevant, and Time-bound (SMART), focusing squarely on the author's modules and providing a clear roadmap for development.

The Research Questions were refined to be action-oriented, focusing on evaluating the effectiveness and impact of the developed solutions. They successfully shift the focus from mere implementation to assessing the tangible benefits in security, efficiency, and user experience.

The Project Research Scope was presented with exceptional clarity using a detailed table. This section is particularly effective as it not only outlines the author's contributions in detail but also summarizes the teammate's modules, providing a holistic view of the entire system while maintaining a strict boundary for the individual project scope. This prevents scope creep and sets realistic expectations.

The Project Milestones and Schedule provides a realistic and well-paced timeline for the project's lifecycle, from proposal to final submission. The use of a Gantt chart and table offers a clear visual representation of the plan, ensuring all critical phases are accounted for.

Finally, the Software and Hardware section accurately identifies all necessary technological tools and platforms required for the successful execution of the project, demonstrating thorough planning in terms of technical resource allocation.

In evaluation, this chapter has effectively achieved its goal of providing a complete introduction and planning framework for the project. It clearly articulates the what, why, and how of the "Vanguard" system. The content is logically structured, well-supported, and sets a strong, clear precedent for the subsequent chapters of detailed analysis, design, and implementation that will follow.

## Chapter 2

# Literature Review

## **2. Literature Review**

### **2.1 Background**

Vanguard (Condominium Management Application) is developed to modernize and digitalize the daily operations of residential communities by replacing manual, paper-based processes with an integrated mobile and cloud-based management platform. Its purpose is to streamline essential activities such as resident management, visitor registration, facility booking, maintenance reporting, and payment handling into one unified and secure ecosystem. By centralizing these functions, the system aims to enhance convenience for residents, improve efficiency for administrative staff, and strengthen overall community governance.

#### **Current Challenges**

Many condominium environments still rely heavily on outdated processes, creating several operational inefficiencies. Manual visitor tracking using physical logbooks often results in incomplete or inaccurate records, delays in identity verification, and increased security risks due to the inability to reliably trace unauthorized access. Facility booking processes face similar challenges, as traditional methods—including phone calls, WhatsApp messages, or physical forms—frequently lead to scheduling conflicts, double-bookings, and unnecessary manual workload for staff who must coordinate approvals and maintain records.

Communication among residents, administrators, and security personnel is also commonly fragmented. Important notices may be missed, and residents lack structured channels to submit feedback or report maintenance issues, resulting in slow response times and reduced satisfaction. Administratively, property management teams bear a heavy workload due to manual billing, document verification, and maintenance coordination tasks, contributing to inefficiency and higher operational costs. Furthermore, the absence of a centralized digital system reduces transparency, making it difficult to access accurate data regarding payments, facility usage, and maintenance activities, ultimately hindering informed decision-making.

#### **Purpose of the System**

The system aims to address these challenges by introducing a mobile-first condominium management platform that consolidates all key operations into a single application. Its purpose is to enhance security, transparency, and operational efficiency through real-time data synchronization and automated workflows. Vanguard also seeks to improve communication and engagement between residents and administrators by offering instant notifications, integrated chat functions, and structured

digital feedback mechanisms. Additionally, the system enables data-driven decision-making by providing administrators with dashboards and analytics tools that offer insight into community activities and operational trends.

### **Technological Foundation**

Vanguard is built on modern technologies that support reliability, scalability, and real-time performance. The Flutter framework enables cross-platform mobile development, ensuring a consistent user experience across both Android and iOS devices. Backend services are powered by Firebase, which provides authentication, real-time database capabilities through Firestore, and secure cloud storage for documents. Artificial Intelligence and Machine Learning technologies, particularly Google ML Kit, enable on-device facial recognition and OCR-based document verification, strengthening security and reducing manual verification effort. QR code technology is used to generate contactless, time-bound visitor passes that enhance traceability and entry control. Cloud Functions and messaging services automate essential workflows—such as booking confirmations and alerts—and ensure seamless communication among residents, administrators, and security personnel.

### **Relevance to Sustainable Development Goals (SDGs)**

The system aligns with key objectives of the United Nations Sustainable Development Goals. Under SDG 10, Vanguard supports reduced inequalities by incorporating accessibility features that cater to elderly and disabled residents, promoting equal participation in digital community management and reducing the technological gap within residential environments. In relation to SDG 11, the application contributes to the development of sustainable cities and communities by promoting secure, efficient, and transparent management practices. Through digital transformation, the system enhances safety, optimizes resource usage, and supports the evolution of smarter urban living environments.

## 2.2 Literature review

### 2.2.1 Sustainable Development Goals (SDGs)



**Figure 2.1 : Sustainable Development Goals (SDGs)**

The United Nations' Sustainable Development Goals (SDGs) provide a global framework for promoting inclusive, safe, resilient, and sustainable communities. In particular, SDG 11 (Sustainable Cities and Communities) aims to make cities inclusive, safe, resilient, and sustainable, while SDG 10 (Reduced Inequalities) focuses on reducing disparities within and among countries by promoting social, economic, and political inclusion (United Nations, n.d.-a; United Nations, n.d.-b). These goals emphasize that urban development and technological interventions should benefit all residents and reduce vulnerability in rapidly urbanizing contexts (UN-Habitat, 2022).

#### **Relevance of SDG 11 to condominium management systems.**

Digital condominium management solutions—by centralizing visitor control, facility scheduling, maintenance workflows, and real-time reporting—can directly support targets under SDG 11 that call for safer and better-managed urban settlements (United Nations, n.d.-b). UN-Habitat highlights that smart urban interventions and resilient digital infrastructure can improve service delivery, safety, and resource use in cities, but also warns that governance and equity issues must be managed to realise these benefits (UN-Habitat, 2022). Therefore, applying mobile-first, cloud-backed systems in residential communities can contribute to more resilient and efficiently managed urban housing, provided that implementation attends to governance, privacy, and inclusivity concerns.

#### **Relevance of SDG 10 to digital condo platforms.**

SDG 10's emphasis on inclusion implies that technology solutions should reduce—not

widen—inequalities (United Nations, n.d.-a). International agencies stress that digital accessibility and inclusive design are essential to ensure persons with disabilities, older adults, and other vulnerable groups can participate fully in digital services—for example, through accessible interfaces, assistive features, and affordability measures (World Bank, 2023). By incorporating accessibility features such as voice guidance, simplified UI, and alternative interaction methods, and by explicitly aiming to close the local digital divide, a condominium management system can support SDG 10’s objectives of social and economic inclusion at the community level (ITU, 2023; United Nations, n.d.-a).

### **2.2.2 Overview of Artificial Intelligence (AI)**

Artificial Intelligence (AI) refers to the development of computer systems capable of performing tasks that typically require human intelligence, such as perception, reasoning, and learning . Over the past decade, AI has evolved into a core driver of digital transformation, enabling organizations to enhance efficiency, automate complex processes, and deliver personalized user experiences. Within the domain of smart community and property management, AI serves as a key enabler of automation, predictive analysis, and secure identity management (Kodydek & Anderl, 2023).

The strategic use of AI technologies in condominium management yields significant advantages in two major areas:

#### **Enhancing Community Security and Verification:**

AI-powered tools such as biometric authentication and intelligent document verification increase accuracy and reduce human error in resident and visitor identification (Entrust, 2025). By automating verification tasks that were traditionally manual, AI enhances the reliability of security screening and minimizes risks of unauthorized access. This aligns with the goal of creating safer and smarter residential communities (UN-Habitat, 2022).

#### **Optimizing Operational Efficiency and Decision-Making:**

AI supports data-driven management by analyzing real-time information on facility usage, payments, and maintenance activities. Automated data processing allows administrators to make informed decisions quickly, improving response times and reducing administrative workload. AI-enabled automation also helps standardize workflows, ensuring greater transparency and consistency in daily operations (Haenlein & Kaplan, 2019).

The adoption of AI in this project is therefore not limited to technological innovation—it plays a strategic role in transforming traditional condominium management into a smart, sustainable, and inclusive ecosystem. By integrating AI into key modules such as resident management, visitor access, and maintenance tracking, the system advances both operational excellence and community

well-being, contributing directly to the realization of SDG 11: Sustainable Cities and Communities (United Nations, n.d.-b).

### **2.2.3 Types of Artificial Intelligence (AI)**

Vanguard (Condominium Management Application) incorporates Applied Artificial Intelligence, primarily focused on Computer Vision and Machine Learning techniques to automate document verification and identity authentication processes. These types of AI fall under the broader categories of Supervised Learning and Deep Learning, leveraging pattern recognition and neural network models for high-accuracy detection.

#### **Computer Vision and Deep Learning**

Computer Vision enables machines to interpret and understand visual information from the physical world through image analysis and feature extraction (Hyperscience, n.d.). Deep learning, particularly Convolutional Neural Networks (CNNs), has become the foundation of modern Computer Vision applications due to its ability to automatically learn hierarchical visual representations (LeCun, Bengio, & Hinton, 2015). CNN models have been successfully applied in image classification, facial recognition, and text extraction, achieving superior accuracy over traditional feature-based methods (Guo & Zhang, 2019).

#### **Optical Character Recognition (OCR) and Document AI**

OCR technology, enhanced by AI-driven neural networks, converts scanned or photographed documents into machine-readable text. Modern AI-based OCR engines—such as Google Document AI and Firebase ML Kit’s Text Recognition API—use deep learning algorithms to identify fonts, detect patterns, and validate textual content (Smith, 2021; Google Cloud, 2023). In this project, OCR is utilized for automated verification of residents’ title deeds, identification cards, and tenancy agreements, ensuring faster onboarding and reduced human error (JurisTech, 2023).

#### **Facial Recognition and Biometric AI**

Facial recognition technology is another critical AI component integrated into the system. It applies deep convolutional neural networks (DCNNs) to analyze and compare facial patterns with stored templates for secure access control. The AI algorithms detect facial landmarks, generate embeddings, and perform matching in real time, offering a contactless, efficient, and reliable method of identity verification for both residents and visitors (Alghamdi et al., 2022).

### **2.2.4 AI in Help & Support Module**

AI technologies are embedded across multiple modules of Vanguard (Condominium Management Application) to enhance automation, accuracy, and user experience. These intelligent components ensure the system operates efficiently while maintaining strong security and inclusivity standards.

### **Resident & Tenant Management Module**

This module integrates OCR-based AI verification to automatically validate residents' property ownership and tenant information. When users upload title deeds or identification documents, the OCR engine—powered by Firebase ML Kit's Text Recognition API—extracts and validates textual data in real time (Google Cloud, 2023). The use of AI significantly reduces manual administrative workload and eliminates fraudulent submissions.

### **Visitor & Security Access Module**

AI-driven facial recognition ensures that only authorized individuals can enter the premises. Using deep learning-based facial embedding models, the system compares live camera inputs against stored templates to verify identity instantly. This implementation allows for touchless entry, increasing both safety and convenience, and is supported by a QR code fallback system for guests without registered facial data (Alghamdi et al., 2022).

## **2.3 Feasibility Studies**

The feasibility study evaluates the practicality of developing and implementing Vanguard (Condominium Management Application), focusing on its technical, operational, and economic aspects. The purpose is to ensure that the system can be effectively built, deployed, and maintained within real condominium management settings.

### **2.3.1 Technical Feasibility**

This section assesses whether the necessary tools, technologies, and expertise are available to successfully implement the system.

**Development Framework:** The system will be developed using Flutter (Dart) for cross-platform mobile application development. This framework enables deployment on Android devices using a single codebase, improving efficiency and reducing maintenance complexity (Google Developers, 2023).

**Backend & Cloud Services:** Firebase will be adopted as the backend solution, providing real-time database synchronization, secure user authentication, and cloud data storage. This ensures a stable and scalable infrastructure that supports multiple concurrent users efficiently (Google Cloud, 2023).

**AI Integration:**

The project integrates AI-driven Computer Vision and Optical Character Recognition (OCR) to automate document verification, while Facial Recognition technology enhances access control and identity security. These modules rely on pre-trained deep learning APIs such as Firebase ML Kit and Google Document AI, which are proven and developer-accessible technologies (JurisTech, 2023).

**Hardware Requirements:** A standard laptop equipped with Flutter SDK and Android Studio, along with an Android smartphone for testing, is sufficient for the development and deployment process. Stable internet connectivity is required for real-time cloud synchronization and AI API integration.

**Conclusion:** The system is technically feasible, as all tools and technologies are stable, well-documented, and readily available for development.

### 2.3.2 Operational Feasibility

Operational feasibility examines whether the system fulfills user needs and can be effectively integrated into the daily operations of condominium management.

**Enhanced Security and Verification:** The integration of AI-powered document scanning and facial recognition improves entry security and eliminates manual identity checks, reducing the risk of unauthorized access (Entrust, 2025).

**Workflow Automation:** Automating facility booking, maintenance requests, and payment collection reduces administrative workload and human error, improving response time and service efficiency (Hyperscience, n.d.).

**User Experience:** A mobile-first design ensures that both residents and administrators can easily access and operate the system through a clean, user-friendly interface, enhancing overall satisfaction and inclusivity.

**24/7 Accessibility:** Cloud-based data storage and automated AI verification allow the system to function continuously without manual supervision, providing real-time access to essential services anytime and anywhere.

**Conclusion:** The system is operationally feasible, as it aligns with the practical workflows of condominium management while improving convenience, security, and responsiveness.

### 2.3.3 Economic Feasibility

Economic feasibility evaluates whether the project's expected benefits justify its development and operational costs.

**Low Development Cost:** The use of open-source and affordable technologies—such as Flutter, Firebase, and Google ML Kit—minimizes financial expenses. Basic API tiers are sufficient for prototype and pilot deployment, making it cost-effective (Google Cloud, 2023).

**Utilization of Existing Resources:** All required hardware and software development tools are readily available, eliminating the need for additional investment in specialized infrastructure.

**Long-Term Value:** By reducing administrative workload, eliminating paper-based processes, and improving operational accuracy, the system provides significant long-term cost savings. Furthermore, enhanced efficiency and resident satisfaction contribute to a stronger community management reputation and sustainability.

**Conclusion:** The project is economically feasible, offering substantial operational and financial benefits relative to its low implementation cost.

## 2.4 Chapter Summary and Evaluation

This chapter has provided an in-depth review of the theoretical and practical foundations supporting the development of Vanguard (Condominium Management Application), emphasizing its relevance to sustainable urban living and modern digital transformation. Through the analysis of related literature and feasibility assessments, the study confirmed that the integration of Artificial Intelligence (AI), Optical Character Recognition (OCR), and Facial Recognition technologies can significantly enhance the efficiency, security, and inclusivity of condominium management systems.

The literature review established the project's alignment with the United Nations Sustainable Development Goals (SDG 10 and SDG 11), demonstrating its role in promoting inclusive digital participation and creating safer, smarter residential communities. The discussion on AI technologies highlighted how Applied AI, Computer Vision, and Machine Learning provide the technical foundation for document verification, identity authentication, and operational automation. These technologies are not only technically mature but also scalable, supporting long-term sustainability and accessibility goals in community management.

The feasibility study further demonstrated that the project is both technically and economically viable. The chosen development stack—Flutter for cross-platform development and Firebase for cloud-based operations—ensures stable performance, scalability, and cost efficiency. The system's design, grounded in AI-driven automation, is operationally practical

for real-world condominium settings. It reduces administrative workload, improves service accuracy, and enhances user experience for residents and management staff alike.

In evaluation, the findings from this chapter confirm that Vanguard(Condominium Management Application) is not only feasible but strategically valuable. It aligns technological innovation with sustainability objectives, addressing modern urban challenges such as security, accessibility, and management transparency. The insights gained from this chapter directly inform the next stage of system design and implementation, ensuring that technical decisions remain consistent with the project's social, operational, and environmental goals.

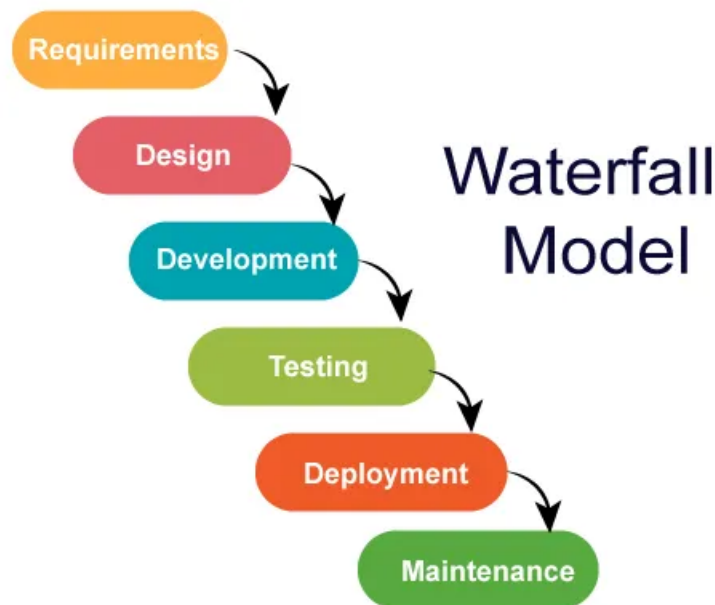
## Chapter 3

# **Methodology and Requirements Analysis**

## 3. Methodology and Requirements Analysis

### 3.1 Methodologies

#### 3.1.1 Waterfall Methodology



**Figure 3.1 : Waterfall Model (Jayathilaka, C., 2020)**

The Waterfall methodology was adopted as the development framework for Vanguard (Condominium Management Application) due to its emphasis on structure, documentation, and linear progression. This model divides the system development process into consecutive stages—beginning with requirement analysis and proceeding through system design, implementation, testing, and finally deployment. Each phase must be fully completed before moving to the next, which ensures a disciplined and traceable workflow throughout the project lifecycle.

The Waterfall approach is particularly suitable for projects with clearly defined goals and stable requirements. In this project, the functional objectives—such as resident management, visitor authentication, facility booking, maintenance request handling, and payment management—were determined at the proposal stage and remained consistent throughout development. This stability made a linear, step-by-step methodology ideal, as it allowed for comprehensive documentation and systematic validation of each module.

From a project management perspective, the Waterfall model provides several advantages that align closely with the needs of an academic final-year project. Its sequential structure supports milestone-based progress tracking, making it easier to plan deliverables and monitor development.

Each stage generates tangible outputs, such as requirement documentation, design diagrams, and test results, which serve as evidence of progress during supervision and evaluation. Furthermore, the model's emphasis on verification at each stage minimizes integration errors and ensures that modules—particularly those involving AI-based verification and facial recognition—are thoroughly tested before deployment.

The methodology was applied in a structured manner across the system's development. During the Requirement Analysis phase, the challenges of traditional condominium management were examined, and key system objectives were established. In the Design phase, user interface wireframes, database structures, and system architecture were developed using Flutter and Firebase as core technologies. The Implementation stage followed a modular development pattern, starting from user authentication and identity verification before moving to booking and payment components. Once implementation was complete, Testing was conducted to verify system reliability, user interaction flow, and AI accuracy. The process concluded with Deployment, where the functional prototype was demonstrated and evaluated by the project supervisor.

Ultimately, the Waterfall methodology proved highly effective for this project. Its linear nature promoted an organized and traceable workflow, while its documentation requirements supported academic assessment. By following this structured development cycle, the project achieved consistency, reliability, and maintainability across all modules—ensuring that the final system met both technical and operational expectations.

### **3.2 Requirement Analysis**

To ensure that Vanguard (Condominium Management Application) accurately meets user expectations and operational needs, a questionnaire-based requirement analysis was conducted. The questionnaire served as the primary data collection method, aiming to gather information from individuals directly involved in condominium operations, including residents, administrative staff, and security personnel.

The survey was designed to identify the key challenges in current condominium management practices, particularly in areas such as visitor verification, facility booking, payment tracking, and maintenance requests. It also aimed to measure user interest and readiness in adopting a mobile-based management system with AI features such as Optical Character Recognition (OCR) and facial recognition.

The questionnaire consisted of both closed-ended and open-ended questions to collect both quantitative and qualitative insights. Closed-ended questions used Likert scales (e.g., *Strongly Agree* to *Strongly Disagree*) to quantify user preferences, while open-ended questions allowed participants to

express specific frustrations or suggestions. This mixed-method approach provided measurable data to support decision-making while capturing user perspectives in greater depth.

Responses were collected from a diverse group of condominium residents and management representatives to ensure a balanced perspective. The collected data highlighted several recurring issues, such as delays in maintenance handling, unclear communication between residents and management, difficulty in accessing real-time payment records, and the need for improved visitor authentication procedures. Based on this analysis, the following system requirements were defined and categorized into Functional and Non-Functional Requirements.

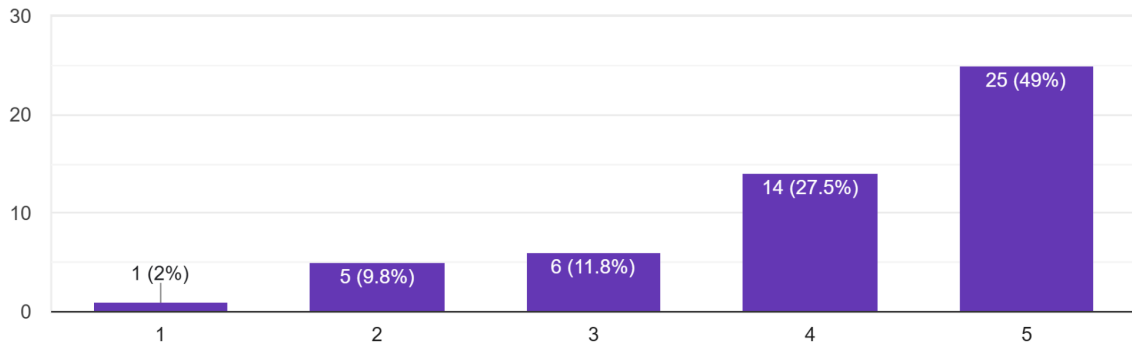
### **3.2.2 Questionnaire**

A targeted questionnaire was conducted to gather quantitative and qualitative insights from condominium residents, tenants, administrative staff, and security personnel, forming a crucial foundation for defining the system requirements of the Vanguard Condominium Management Application. The feedback collected from participants highlighted several recurring operational challenges within the existing manual processes, including inefficiencies in visitor verification, frequent miscommunication between residents and management, and a lack of transparency in facility booking and resident-tenant record handling. The questionnaire further revealed strong interest among users in adopting a centralized mobile platform equipped with modern features such as QR-based visitor passes, real-time facility scheduling, document OCR verification, and facial recognition for secure access. With 51 valid responses, the data showed clear trends that guided the system design—particularly the need to digitalize resident onboarding, strengthen condominium security measures, and streamline facility management workflows. These findings served as empirical justification for the development of the Resident & Tenant Management Module, Visitor & Security Access Module, and Facility Booking Module, ensuring that the final system directly addresses community-validated needs and user expectations.

## 1. Resident & Tenant Management Module

4. I find the current resident or tenant registration process slow or inconvenient.

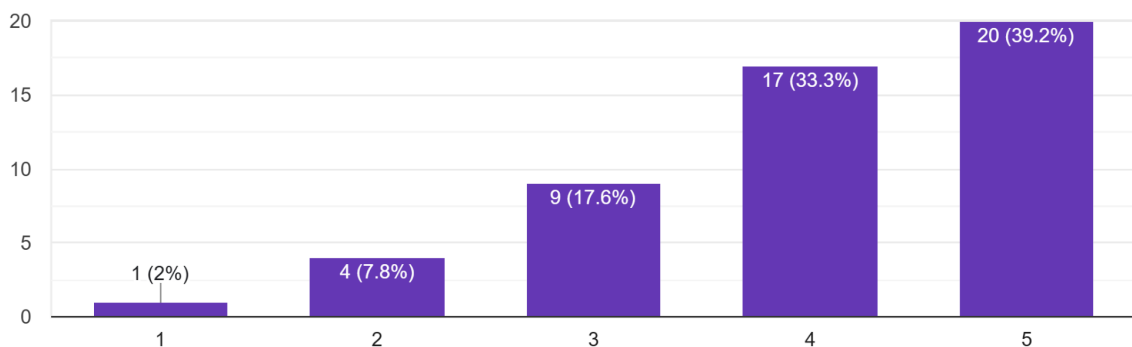
51 responses



**Figure 3.2.2.1 : Survey Results – Resident & Tenant Registration Difficulties**

5. Uploading documents manually (e.g., title deed, tenancy agreement) is troublesome.

51 responses

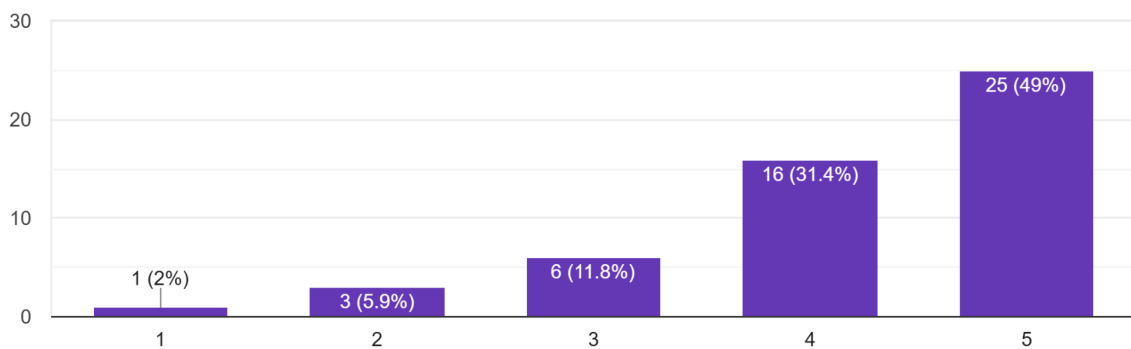


**Figure 3.2.2.2 : Survey Results – Resident & Tenant Registration Difficulties**

The survey results show that a significant proportion of respondents encounter inconvenience with the current resident or tenant registration process. A combined 76.5% of participants selected “Agree” (27.5%) or “Strongly Agree” (49%), indicating widespread dissatisfaction with manual registration workflows that typically involve paper forms, repetitive submissions, and slow administrative processing. Only 2% strongly disagreed, suggesting that negative experiences with the existing system are nearly universal. This trend highlights a clear need for a streamlined, centralized onboarding mechanism to reduce redundancy and improve efficiency for both residents and management staff.

Similarly, when asked about the challenges of manually uploading documents such as title deeds or tenancy agreements, over 72% of respondents expressed agreement or strong agreement. With 33.3% agreeing and 39.2% strongly agreeing, it is evident that current document submission methods—often involving email, WhatsApp, or physical paperwork—create inconvenience, delays, and verification inconsistencies. These findings emphasize the necessity of implementing a digitalized onboarding framework within the Vanguard application. By integrating automated OCR-based document extraction and secure digital submission, the proposed Resident & Tenant Management Module directly addresses the inefficiencies highlighted by participants, ensuring a faster, more reliable, and user-friendly verification process.

6. I would prefer a digital verification system that uses OCR to verify documents automatically.  
51 responses



**Figure 3.2.2.3 : Survey Results – Preference for OCR-Based Digital Verification**

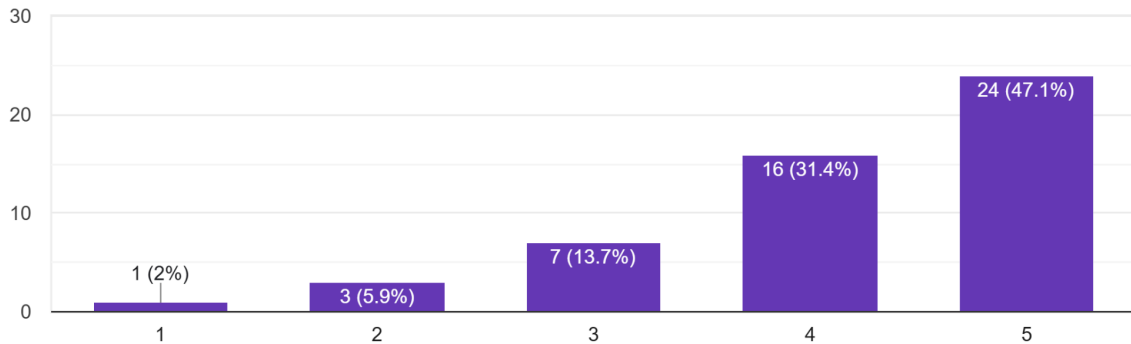
The survey results show overwhelmingly strong support for the implementation of an automated digital verification system using OCR technology. A substantial 49% of respondents selected “Strongly Agree”, with an additional 31.4% selecting “Agree”, demonstrating that over 80% of participants prefer a digital solution over traditional manual verification methods. Only a very small minority (approximately 2% selecting “1” and 5.9% selecting “2”) expressed resistance toward the idea. This significant skew toward the higher end of the scale highlights residents’ readiness to adopt AI-enabled document processing as a means to reduce administrative delays and eliminate the repetitive, error-prone nature of manual verification workflows.

These findings directly validate the decision to integrate OCR document extraction using Google ML Kit within the Vanguard Resident & Tenant Management Module. By automating the reading and verification of identity documents, title deeds, and tenancy agreements, the system can significantly streamline the onboarding process, reduce mistakes, and minimize dependency on manual administrative review. The strong positive reception shown in the survey suggests that users not only understand the benefits of digital verification but are also eager to transition away from outdated methods. This empirical evidence reinforces OCR integration as a necessary and user-driven feature that enhances efficiency, accuracy, and user satisfaction.

## 2. Visitor & Security Access Module

9. The current visitor registration and tracking process (logbook/WhatsApp/etc.) is inefficient.

51 responses



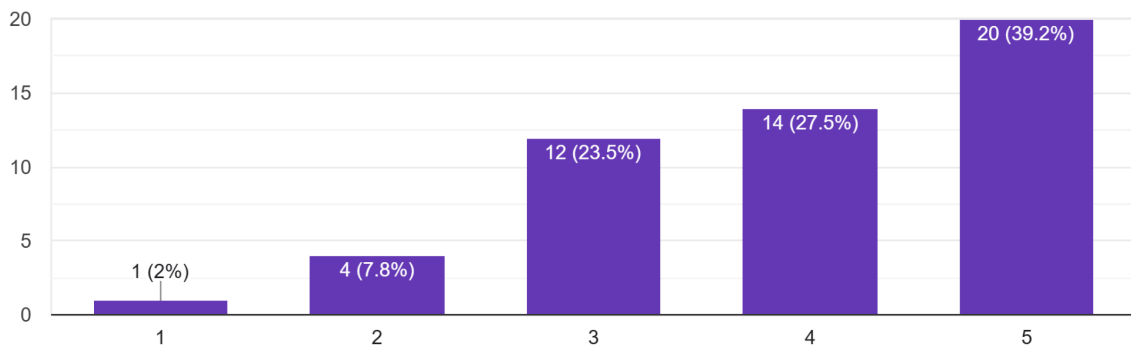
**Figure 3.2.2.4 : Survey Results – Visitor Registration and Tracking Inefficiencies**

The survey results indicate a clear dissatisfaction with the current visitor registration and tracking process used within the condominium. An overwhelming 47.1% of respondents strongly agreed and 31.4% agreed that existing methods—typically involving handwritten logbooks, WhatsApp messages, or informal communication—are inefficient. Only a very small minority expressed disagreement. The distribution strongly suggests that residents perceive the visitor management workflow as outdated, unstructured, and prone to errors. These fragmented methods not only slow down visitor processing at the guardhouse but also create significant security gaps, as handwritten logs are difficult to validate, track, or audit. The data highlights a pressing need for a more reliable and systematic approach to managing visitor information.

These findings reinforce the necessity of implementing a centralized, automated Visitor & Security Access Module within Vanguard. By incorporating QR-based visitor passes, digital pre-registration, and real-time verification features, the system directly addresses the inefficiencies identified in the survey. With Firebase integration ensuring accurate logging and accessible visitor records, Vanguard offers a more secure, transparent, and efficient alternative to the traditional logbook-based workflow. The strong consensus among respondents validates the adoption of digital visitor management as a core feature of the system.

10. I am concerned about unauthorized visitors entering the condominium.

51 responses



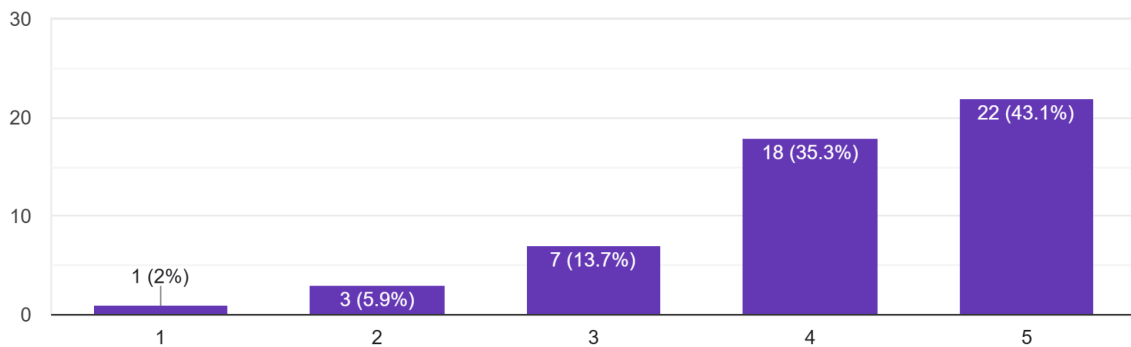
**Figure 3.2.2.5 : Survey Results – Concerns About Unauthorized Visitor Entry**

A substantial proportion of respondents expressed concern regarding unauthorized visitors entering the condominium. 39.2% strongly agreed and 27.5% agreed, while only minimal disagreement was reported. This indicates that security and access control are major priorities for the majority of residents. The responses suggest that current procedures fail to provide adequate reassurance, primarily due to manual visitor verification methods that lack identity validation and are easily bypassed. The presence of informal or inconsistent security practices likely contributes to residents' anxiety about safety and access control within the community.

These results underscore the importance of enhancing condominium security through automation and identity verification. Vanguard addresses this concern through features such as QR code authentication, facial recognition for resident entry, and digital visitor logs, enabling more reliable tracking and identity confirmation at security checkpoints. By reducing reliance on subjective manual checks, the system strengthens entry control and provides residents with greater confidence in the safety of their living environment. The survey data clearly supports the implementation of a digital access solution to mitigate unauthorized entry risks.

11. A QR-based visitor pass system would make the entry process safer and more convenient.

51 responses



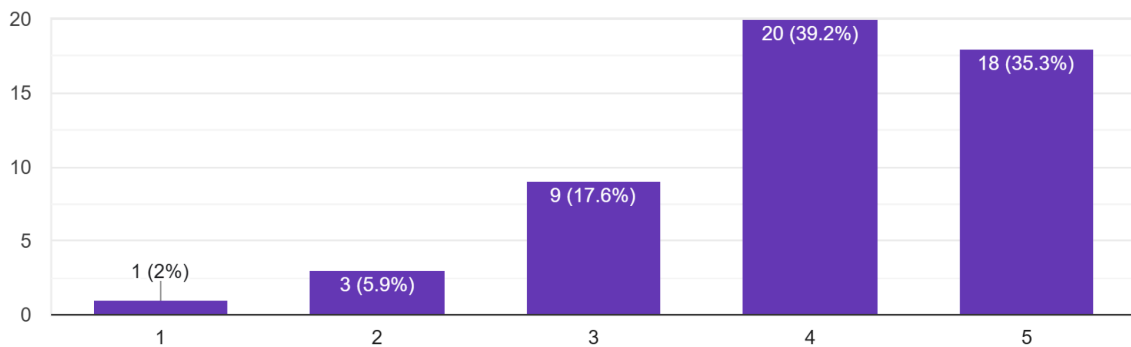
**Figure 3.2.2.6 : Survey Results – Perceived Usefulness of QR-Based Visitor Pass System**

The results indicate strong support for adopting a QR-based visitor pass system, with 43.1% of respondents selecting “Strongly Agree” and 35.3% selecting “Agree.” This means that nearly 80% of participants believe that QR authentication would make the visitor entry process safer and more convenient. Only a negligible percentage expressed disagreement. This overwhelming positive response suggests that residents recognize the limitations of traditional logbook-based processes, which are slow, error-prone, and offer little security. The enthusiasm toward QR passes reflects a growing preference for digital automation and contactless solutions commonly used in modern access control systems.

These findings validate the integration of QR-based visitor registration within the Vanguard system. By generating time-bound, cryptographically signed QR codes for guests, Vanguard directly addresses the inefficiencies identified in the existing workflow. A QR-enabled access mechanism not only improves processing speed at the guardhouse but also enhances traceability, reduces unauthorized entry risk, and offers a convenient experience for both residents and visitors. The strong user acceptance indicated in the survey confirms that QR visitor authentication is a highly justified and community-endorsed feature.

## 12. Facial recognition could improve resident entry security.

51 responses

**Figure 3.2.2.7 : Survey Results – Perceived Security Benefits of Facial Recognition**

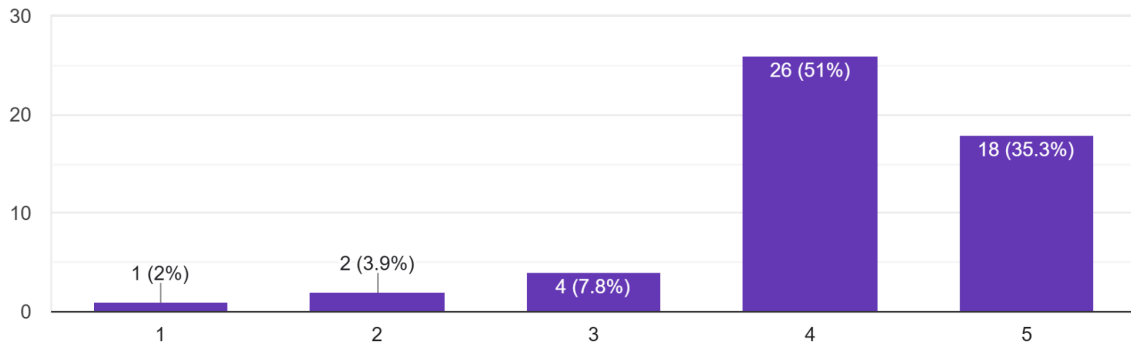
Survey responses reveal that residents are highly receptive to using facial recognition to enhance condominium entry security. A combined 74.5% of respondents agreed or strongly agreed that facial recognition could improve the security of resident entry, with 39.2% selecting “Agree” and 35.3% selecting “Strongly Agree.” Only minimal disagreement was noted. These results indicate a strong trust in biometric authentication as a reliable method for preventing unauthorized access and reducing dependency on traditional methods such as access cards or manual verification, which are often lost, shared, or misused.

These results strongly support the inclusion of a facial recognition component in the Visitor & Security Access Module of Vanguard. By leveraging Google ML Kit’s on-device facial recognition, the system enables secure, hands-free resident entry while maintaining privacy through local processing. The feature directly addresses residents’ security concerns identified in the survey—particularly risks associated with unauthorized entry and inconsistent guardhouse checks. The high acceptance level demonstrates that residents are ready for biometric-enhanced security, making facial recognition an essential and well-supported feature within the system design.

### 3. Facility Booking Module

16. Real-time facility availability would improve my booking experience.

51 responses



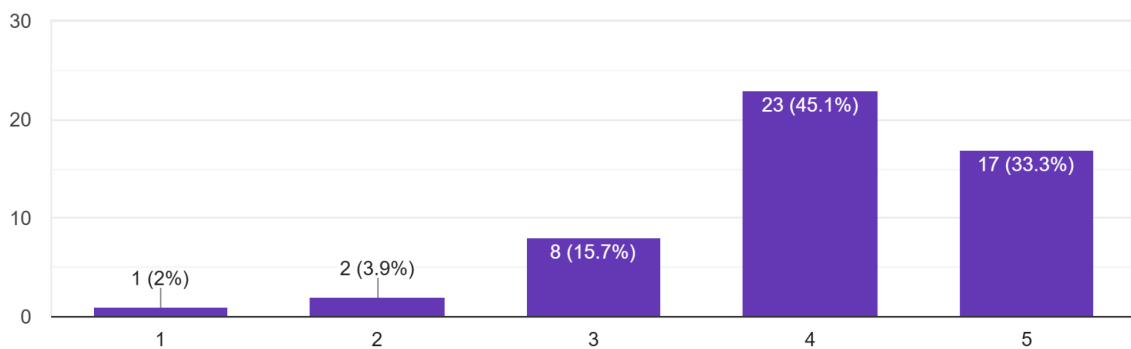
**Figure 3.2.2.8 : Survey Results – Demand for Real-Time Facility Availability**

The survey results show exceptionally strong support for the implementation of real-time facility availability within the condominium booking system. A majority of respondents expressed clear agreement, with 51% selecting “4 – Agree” and 35.3% selecting “5 – Strongly Agree,” indicating that over 86% of participants believe real-time availability would significantly improve their booking experience. Only a very small group expressed disagreement. This overwhelming consensus suggests that residents face recurring frustrations under the current facility booking arrangements, which may rely on manual confirmation, delayed updates, or inconsistent record-keeping. Such limitations often lead to scheduling conflicts, misunderstandings, or wasted trips to fully occupied facilities.

These findings strongly justify the integration of a fully digitalized, real-time scheduling interface within the Vanguard Facility Booking Module. By synchronizing availability data through Firebase and enforcing conflict-prevention logic, the system directly addresses the inefficiencies highlighted by users. Real-time visibility into open and occupied time slots not only improves convenience but also enhances fairness, ensuring that all residents gain equal access to shared amenities. The survey results clearly demonstrate that this feature aligns with user expectations for transparency, responsiveness, and reliability in facility management.

## 17. Automated notifications (booking approval, reminders, etc.) would be helpful.

51 responses

**Figure 3.2.2.9 : Survey Results – Usefulness of Automated Booking Notifications**

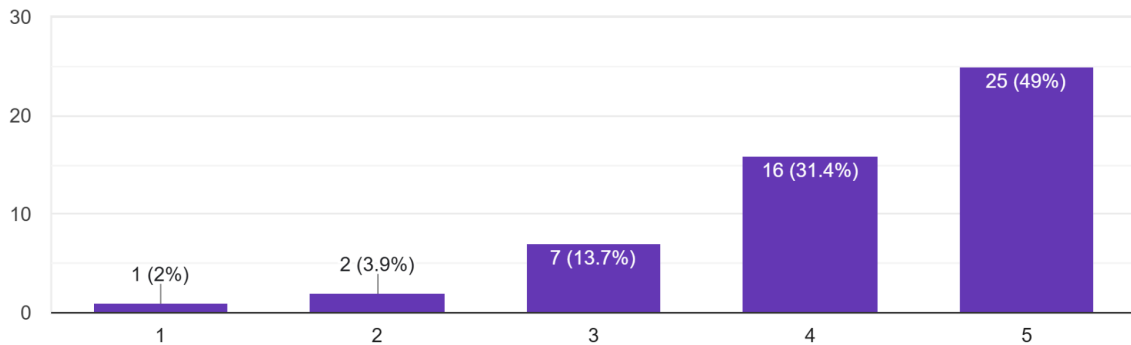
The survey results also highlight the strong user demand for automated notifications related to facility bookings. A combined 78.4% of respondents agreed or strongly agreed that automated updates—such as booking approvals, reminders, and status changes—would be beneficial. Specifically, 45.1% agreed and 33.3% strongly agreed, while only minimal disagreement was reported. This indicates that residents value proactive communication and appreciate systems that help them stay informed without needing to contact management or manually track their bookings. The findings suggest that inconsistent or delayed communication within the current manual process contributes to missed bookings, confusion about approval status, and overall dissatisfaction.

This feedback directly supports the inclusion of automated push notifications through Firebase Cloud Messaging in the Vanguard system. By delivering instant alerts for booking confirmations, cancellations, reminders, and updates, the system enhances clarity and reduces uncertainty for residents. Such automation not only streamlines administrative workload but also improves user engagement and satisfaction by providing timely and transparent communication. The strong endorsement shown in the responses confirms that automated notifications are an essential and user-driven component of an effective condominium facility booking solution.

### System Adoption and AI Readiness

19. I am comfortable using mobile applications for condominium-related tasks.

51 responses



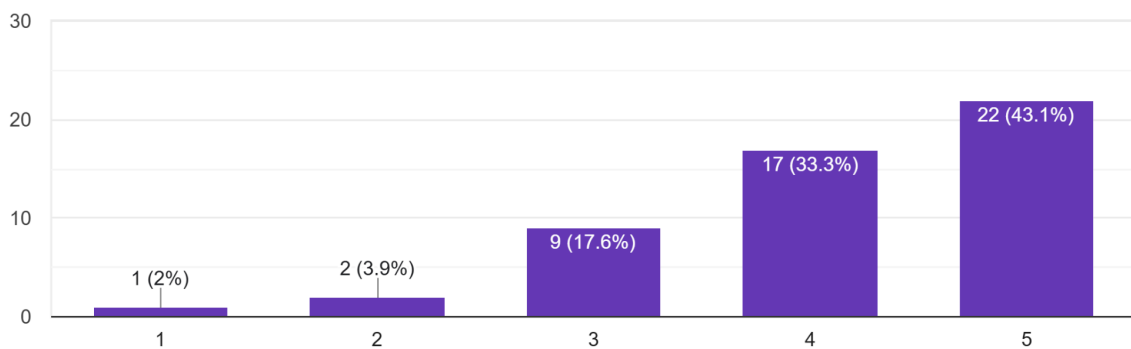
**Figure 3.2.2.10 : Survey Results – Comfort with Mobile Applications for Condominium Tasks**

The survey results show strong user readiness for adopting a mobile-based condominium management system. A combined 80.4% of respondents selected “4 – Agree” or “5 – Strongly Agree,” indicating that the majority of residents and tenants feel comfortable performing condominium-related tasks through mobile applications. Only a very small fraction expressed discomfort, with 2% selecting “1” and 3.9% selecting “2.” This suggests that digital platforms are widely accepted among condominium communities, and mobile applications are increasingly seen as a convenient and familiar tool for daily interactions. The high level of comfort reported by respondents demonstrates that transitioning from manual processes to mobile-based workflows will encounter minimal resistance and aligns with current user behavior trends.

These findings directly support the implementation of Vanguard as a mobile-first solution. Since mobile adoption readiness is already high, features such as digital onboarding, facility booking, visitor pre-registration, and push notification-based communication are positioned to achieve immediate user acceptance. The results also validate the decision to prioritize usability and intuitive design in the system interface, ensuring accessibility for users of varying technological proficiency. The strong positive reception toward mobile usage indicates that a centralized digital application will effectively streamline communication, increase workflow efficiency, and reduce reliance on scattered traditional channels.

20. I am open to using AI features such as OCR document verification and facial recognition.

51 responses



**Figure 3.2.2.11 : Survey Results – Openness to AI-Driven Features (OCR & Facial Recognition)**

Responses regarding AI readiness show similarly strong acceptance among users. A combined 76.4% of participants agreed or strongly agreed that they are open to using AI-powered features such as OCR-based document verification and facial recognition for identity authentication. Among these, 43.1% selected “Strongly Agree,” demonstrating a high level of enthusiasm toward adopting advanced automation technologies. Only a small proportion expressed hesitation, with fewer than 6% selecting the lower end of the scale. This trend indicates that residents recognize the value of AI in improving accuracy, reducing manual verification effort, and enhancing overall condominium security.

These findings strongly justify the integration of AI capabilities within Vanguard’s Resident & Tenant Management Module and Visitor & Security Access Module. OCR technology directly addresses user frustration with manual document handling, while facial recognition enhances resident entry security by minimizing unauthorized access risks. The high acceptance levels shown in the survey confirm that residents are prepared for AI-driven enhancements and perceive them as beneficial rather than intrusive. This readiness provides a strong foundation for implementing intelligent automation within the system, reinforcing the feasibility and relevance of AI integration in condominium management workflows.

### **3.2.2 Functional Requirements**

Functional requirements define the essential operations that the system must perform to achieve its intended objectives and support core condominium management activities.

#### **Resident & Tenant Management Module**

The Resident & Tenant Management Module enables residents and tenants to register and update their personal information through the mobile application, ensuring a seamless onboarding process. The system incorporates AI-based Optical Character Recognition (OCR) to automatically verify uploaded identity documents such as ID cards and tenancy agreements, reducing manual verification workload and enhancing data accuracy. Administrators are provided with dedicated approval controls to validate user information before granting access to the system. All verified personal and tenancy data is securely stored in Firebase Firestore to support reliable record-keeping and future reference.

#### **Visitor & Security Access Module**

The Visitor & Security Access Module implements facial recognition technology to support real-time visitor verification and secure access authorization. Residents can pre-register visitors and generate unique QR codes that serve as temporary entry passes. Each visitor entry is logged with timestamps and verification results, enabling administrators to review access histories and maintain comprehensive security records. Security personnel can track visitor status, manage entry permissions, and ensure that only authorized individuals gain access to the premises.

#### **Facility Booking Module**

The Facility Booking Module provides residents with a real-time interface to view facility availability and current booking statuses. Through the mobile application, users can reserve shared facilities such as halls, gyms, or pools, with built-in conflict prevention logic ensuring that overlapping bookings do not occur. The system automatically sends booking confirmations and reminder notifications to enhance user convenience. Administrators are empowered to approve, modify, or cancel booking requests when necessary, ensuring effective management of shared community resources.

### **3.2.3 Non-Functional Requirements**

Non-functional requirements specify the quality standards and operational constraints that ensure the reliability, usability, security, and overall performance of the system.

#### **Performance Requirements**

The system must be capable of supporting concurrent user access while maintaining minimal latency, with key operations expected to respond within three seconds. Firebase services should provide real-time data synchronization and updates across multiple devices to ensure a smooth and uninterrupted user experience.

#### **Usability Requirements**

The mobile interface is designed to be intuitive and accessible for users of all age groups, including elderly residents. Navigation must remain simple and consistent, supported by clear icons and labels. All forms and data entry components are expected to be responsive and easy to complete, ensuring that users can interact with the system efficiently and without confusion.

#### **Security Requirements**

All users are required to authenticate through Firebase Authentication before accessing any module within the system. Sensitive information—such as facial images, identity documents, and personal data—must be securely encrypted within Firebase Cloud Storage. The system must also enforce strict access control policies to prevent unauthorized access and protect confidential information.

#### **Reliability Requirements**

To ensure data integrity, all information stored in the system must remain accurate and consistent even when accessed or updated by multiple users simultaneously. Firebase's cloud infrastructure should automatically back up data to prevent loss in the event of system failure. Additionally, AI verification models, including OCR and facial recognition, are expected to maintain an accuracy rate of at least 90% under normal operating conditions.

#### **Maintainability Requirements**

The system should be developed with a modular and well-documented codebase to facilitate maintenance and future enhancements. Updates must be deployable without interrupting existing data or disrupting active user sessions. The underlying database structure must be scalable and flexible enough to accommodate potential future expansions, such as integration with IoT devices or additional smart-community features.

### 3.3 Tools

The development of Vanguard (Condominium Management Application) utilizes a range of modern tools and technologies that support efficient mobile application development, testing, and deployment. These tools collectively ensure that the system is stable, scalable, and capable of delivering a secure and seamless user experience. Each tool plays a specific role within the development workflow, from frontend implementation to backend management and system monitoring.

#### 3.3.1 Mobile Application Development

The **Flutter SDK** serves as the primary framework for building the mobile application, enabling cross-platform development using a single codebase to reduce development time and maintenance effort. It provides a wide collection of pre-built widgets that support responsive and visually appealing UI design, while its “hot reload” feature allows developers to instantly preview and test code changes, significantly improving debugging efficiency.

The **Dart programming language** is used as the core language for Flutter. Its support for object-oriented programming and asynchronous operations makes it well suited for handling real-time data interactions throughout the system. Dart also integrates seamlessly with Firebase services and Google ML Kit, ensuring smooth communication between the frontend and backend.

**Firebase** functions as the Backend-as-a-Service (BaaS) platform for the project, offering essential modules such as Firebase Authentication for secure user login and identity management, Cloud Firestore for handling user data, booking information, and transaction histories in real time, and Cloud Storage for storing uploaded documents and images used in OCR verification. Firebase’s built-in scalability, availability, and automatic synchronization eliminate the need for a dedicated backend server and simplify overall system maintenance.

**Android Studio** is used as the main Integrated Development Environment (IDE) for Flutter development. It provides robust tools for device emulation, performance profiling, and testing. Its strong compatibility with Flutter SDK, Firebase plugins, and Gradle-based build automation supports a seamless workflow, while Git integration enables effective version control and collaboration.

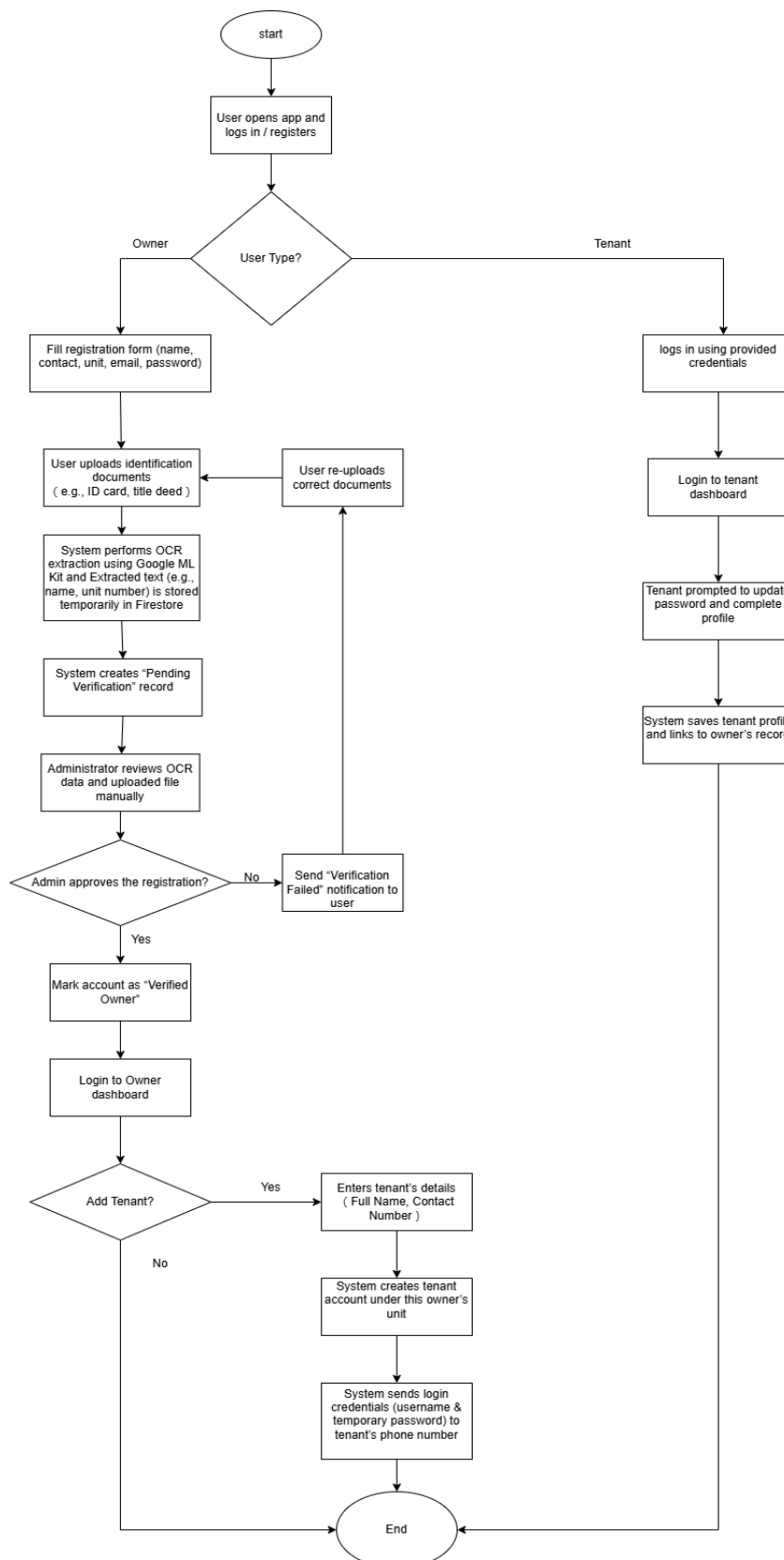
**Google ML Kit** is integrated into the application to support Optical Character Recognition (OCR) and facial recognition functionalities. The Text Recognition API enables the extraction of text from identity documents during registration and verification, while the Face Detection API facilitates real-time facial pattern recognition for secure access control and visitor authentication. ML Kit supports both on-device and cloud-based processing, giving developers the flexibility to balance accuracy and performance.

The **Google Cloud Console** is used to configure and monitor all Firebase and ML Kit resources. It provides access to database analytics, performance logs, and usage statistics, helping the development team track system behavior and optimize performance. It also allows developers to configure authentication methods, API access rules, and data security settings, making it the central platform for backend management.

Finally, **Draw.io** is employed to create system flowcharts, use case diagrams, and database structure diagrams. Its intuitive drag-and-drop interface facilitates effective visualization of workflows and system relationships. It also supports exporting diagrams into image or PDF formats, making it convenient to integrate visuals into documentation and academic presentations.

### 3.4 Flowchart Diagram

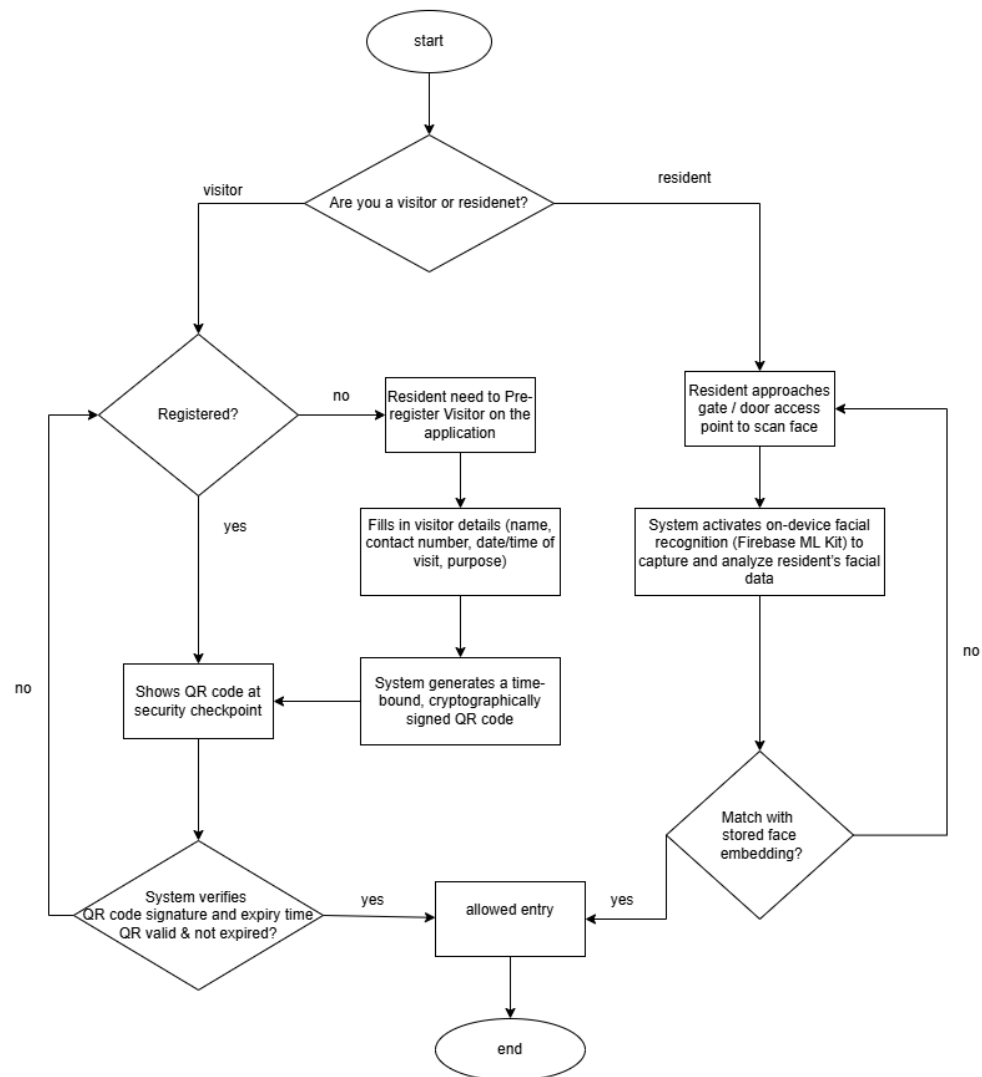
#### Resident & Tenant Management Module



This flowchart outlines the complete registration and login process for residents, covering both property owners and tenants. The flow begins when a user opens the application and selects their account type. Owners are required to provide personal details and upload supporting documents such as title deeds. The system extracts key information using OCR and forwards the submission to the administrator for verification. If approved, the owner's account is activated, allowing them to log in and access the system. Verified owners also have the option to add tenants by submitting their basic details, after which the tenant receives login credentials.

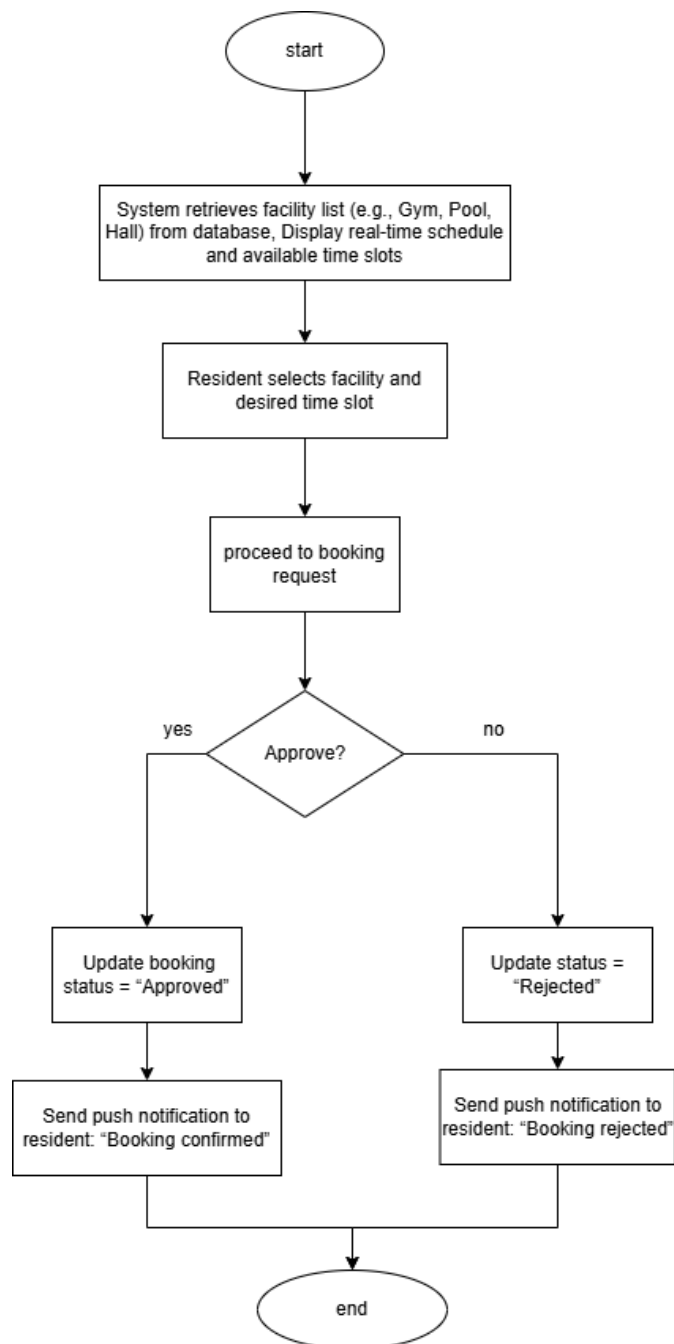
Tenants, upon receiving their credentials, proceed directly to the login process. Once logged in, tenants can view their profiles and access the features assigned to their role. If any owner's document is rejected, the flow returns to the re-upload stage, prompting them to resubmit valid documentation. This structured process ensures secure identity validation, controlled tenant onboarding, and accurate maintenance of resident records within the system.

## Visitor & Security Access Module



This flowchart outlines the access control process for both visitors and residents within the Vanguard system. Visitors must first be pre-registered by a resident, who submits their details through the application. Once registered, the system generates a time-bound, cryptographically signed QR code. Upon arrival, the visitor presents the QR code at the security checkpoint, where the system verifies its validity and expiry before granting entry.

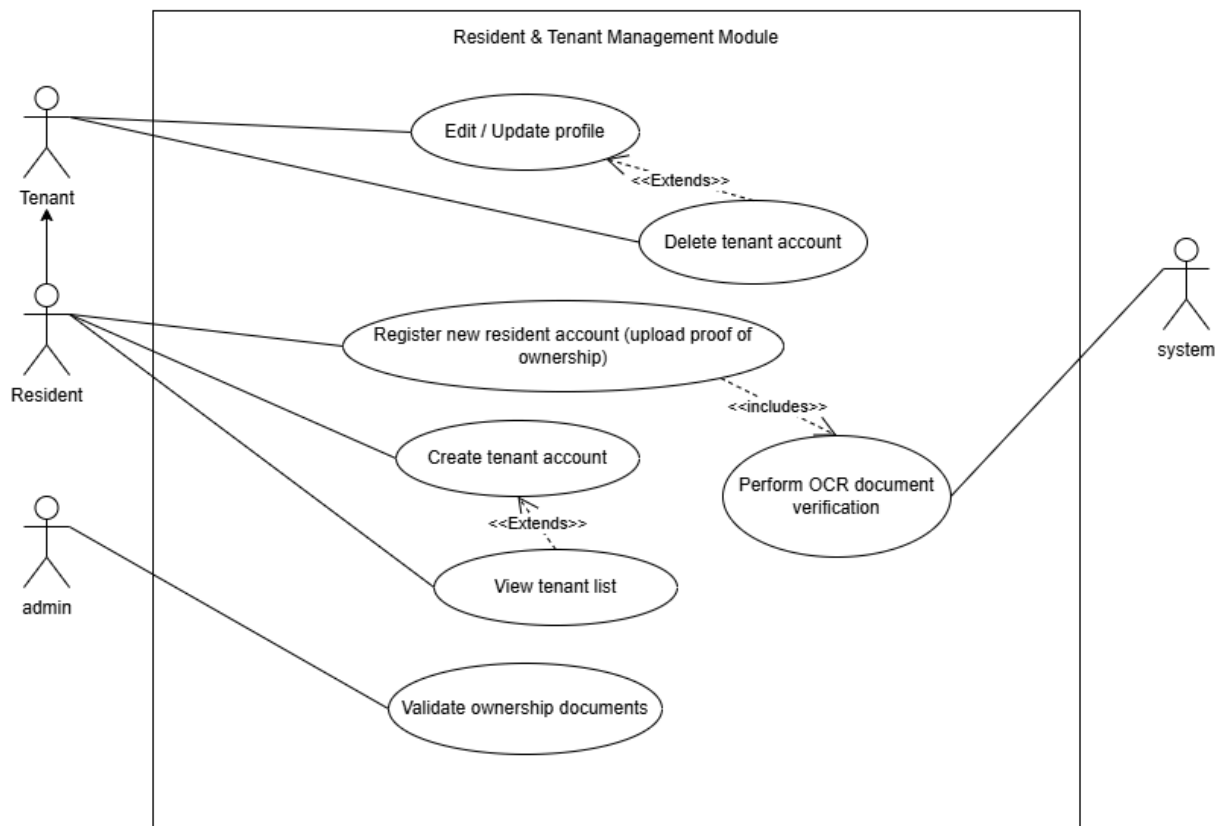
For residents, the system uses on-device facial recognition to authenticate identity at the gate or door access point. The resident's facial data is scanned and compared with their stored face embedding, and access is approved only if the match is successful. This combined approach provides a secure, streamlined, and contactless entry process for all users.

**Facility Booking Module**

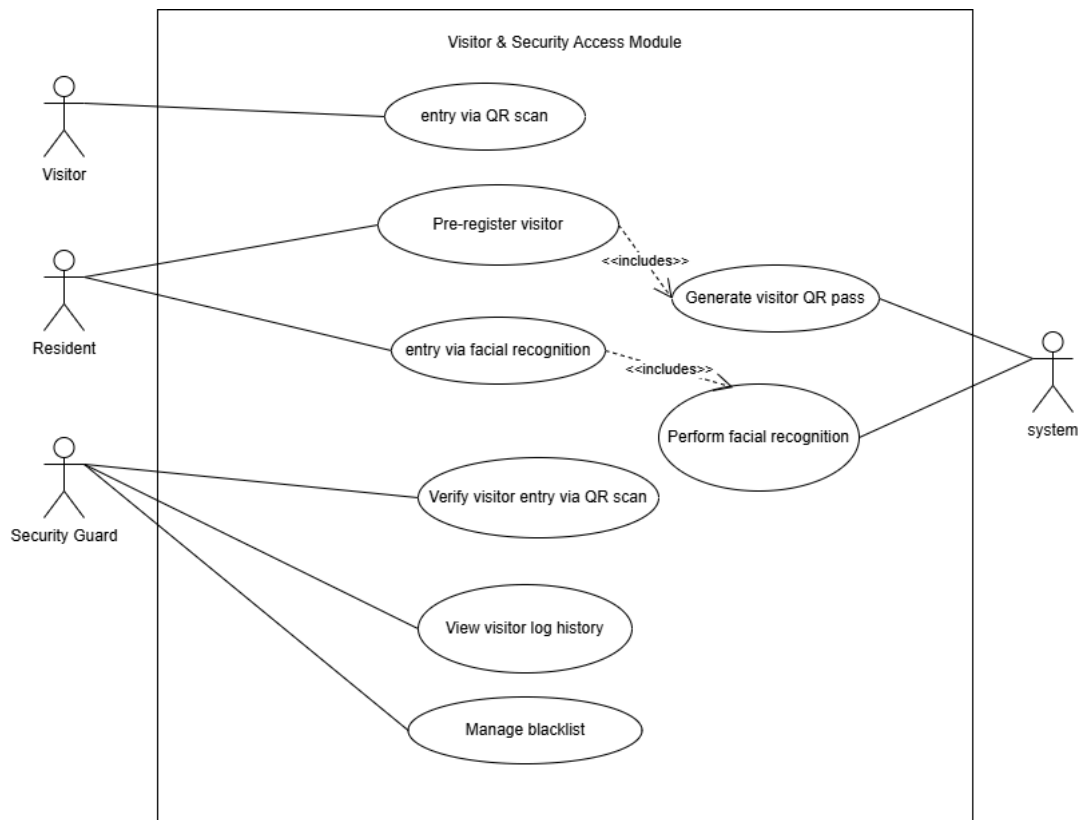
This flowchart illustrates the facility booking process within the Vanguard system. The system first retrieves the list of available facilities and their real-time schedules from the database. The resident selects a facility and a preferred time slot, after which the booking request is submitted for approval. If the request is approved, the system updates the booking status to “Approved” and sends a confirmation notification to the resident. If it is rejected, the status is updated to “Rejected,” and a rejection notification is sent. This process ensures clear, efficient, and transparent facility booking management.

### 3.5 Use Case Diagrams

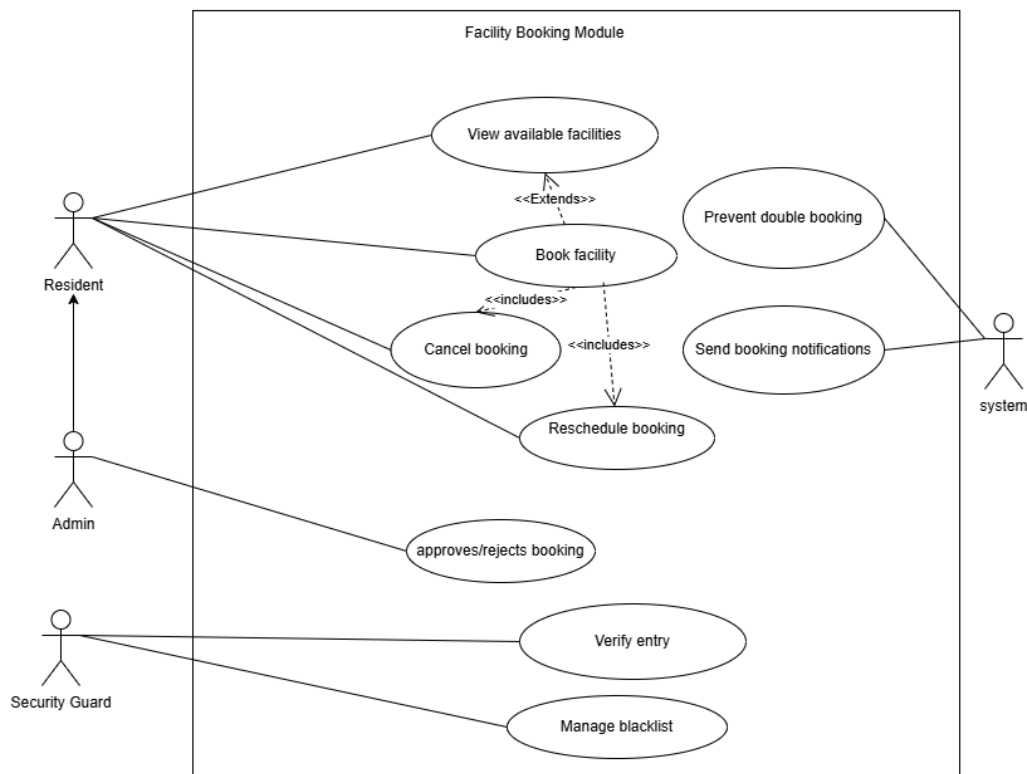
#### Resident & Tenant Management Module



This use case diagram represents the core functionalities of the Resident & Tenant Management Module in the Vanguard system. Residents can register new accounts by uploading proof of ownership, create tenant accounts, and view their list of registered tenants. The system supports OCR-based document verification as an included process to validate ownership documents before approval. Tenants are able to edit or update their profiles, while resident users may also delete tenant accounts when necessary. Administrators are responsible for validating ownership documents submitted during registration. Together, these use cases outline a secure and structured workflow for managing residents and tenants within the condominium environment.

**Visitor & Security Access Module**

This use case diagram illustrates the core processes within the Visitor & Security Access Module. Residents can pre-register visitors, allowing the system to generate a unique QR pass for entry. Visitors use this QR code to access the condominium, while residents themselves may enter through facial recognition, which the system performs as part of an included process. Security guards verify visitor entries by scanning QR codes, view detailed visitor log histories, and manage a blacklist to prevent unauthorized access. Together, these use cases define a secure and streamlined access control workflow for both visitors and residents.

**Facility Booking Module**

This use case diagram outlines the key functionalities of the Facility Booking Module. Residents can view available facilities and submit booking requests, with options to cancel or reschedule their bookings as needed. The system supports automatic conflict prevention to avoid double bookings and sends booking notifications to keep residents informed. Administrators play a supervisory role by approving or rejecting booking requests submitted by residents. Security guards verify entry for booked facilities and manage the blacklist to ensure authorized access only. Together, these use cases support an efficient, transparent, and well-coordinated facility booking process.

## 3.6 Chapter Summary and Evaluation

### Summary

Chapter 3 presents the overall methodology and requirements analysis for the Condominium Management Application. The Waterfall model was selected as the main development approach due to its clear, sequential structure, which ensures that each development phase is properly completed and documented before moving on to the next. This model fits well with the project's fixed objectives, defined module scope, and academic milestones.

A questionnaire-based approach was employed to collect user requirements from condominium residents, tenants, and management staff. This method provided insight into user expectations and system needs related to security, facility booking, and resident management. The responses were analyzed to derive the functional and non-functional requirements, ensuring that the system design emphasizes usability, automation, and data security.

This chapter also details the tools and technologies used, including Flutter, Dart, Firebase, Android Studio, Google ML Kit, and Draw.io, which were selected for their integration capabilities, scalability, and ease of deployment. Each tool plays a vital role in ensuring that the system is responsive, efficient, and capable of handling real-time data processing securely.

### Evaluation

Overall, Chapter 3 establishes a solid foundation for the development of the Condominium Management Application. The chosen methodology promotes structured progress, transparency, and traceability throughout all project phases. The requirement analysis effectively captures real-world user needs, while the selected technologies provide a balance between technical feasibility and system reliability.

A minor limitation of this chapter is that the data collection method primarily relies on questionnaires, which may not fully capture qualitative insights such as user satisfaction or behavioral responses. Nonetheless, the chapter provides a comprehensive framework that guides the system's design, implementation, and evaluation, ensuring that the final product aligns with its intended goals of efficiency, security, and community inclusivity.

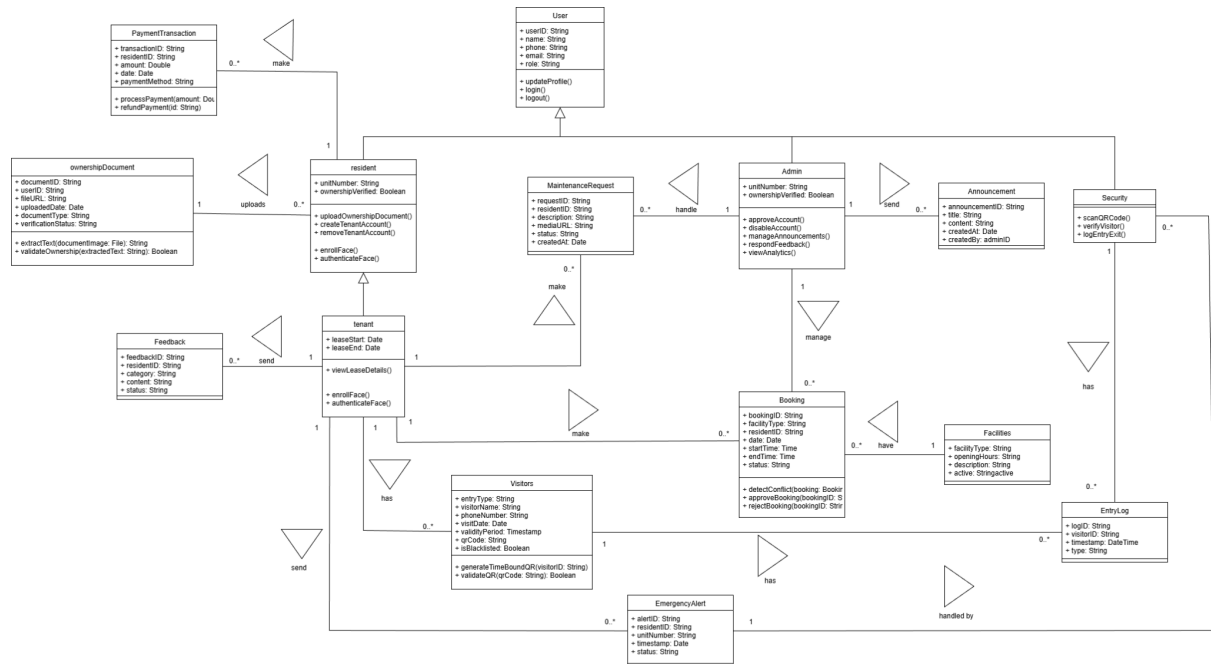
# Chapter 4

## **System Design**

## 4. System Design

## 4.1 UML Diagrams

### 4.1.1 Class diagram



The class diagram illustrates the complete structural architecture of the Condominium Management System by defining the system's core entities, their attributes, behaviors, and the relationships that connect them. It integrates all major functional modules including Resident & Tenant Management, Visitor & Security Access, Facility Booking, Maintenance & Services, Admin Management, and Payment Processing into a single cohesive model. This provides a clear visualization of how the system maintains data consistency, enforces business rules, and enables role-based interactions.

At the center of the diagram is the abstract User class, which establishes common properties such as user identity, contact information, and role designation. Specialized user types Resident, Tenant, Admin, and Security extend this parent class to include role-specific attributes and operations. For example, Residents can upload ownership documents and manage tenants, while Security users verify visitors and log entry activities, and Admins handle approvals, announcements, and system-wide management tasks.

Key operational modules are represented through dedicated entity classes. The ownershipDocument, MaintenanceRequest, Feedback, Booking, Visitors, Facilities, EmergencyAlert, and PaymentTransaction classes store essential data and support CRUD operations for daily condominium activities. Each of these entities is associated with the appropriate user roles through clear

one-to-many or many-to-one relationships, such as residents making bookings, admins handling maintenance requests, or visitors being linked to entry logs.

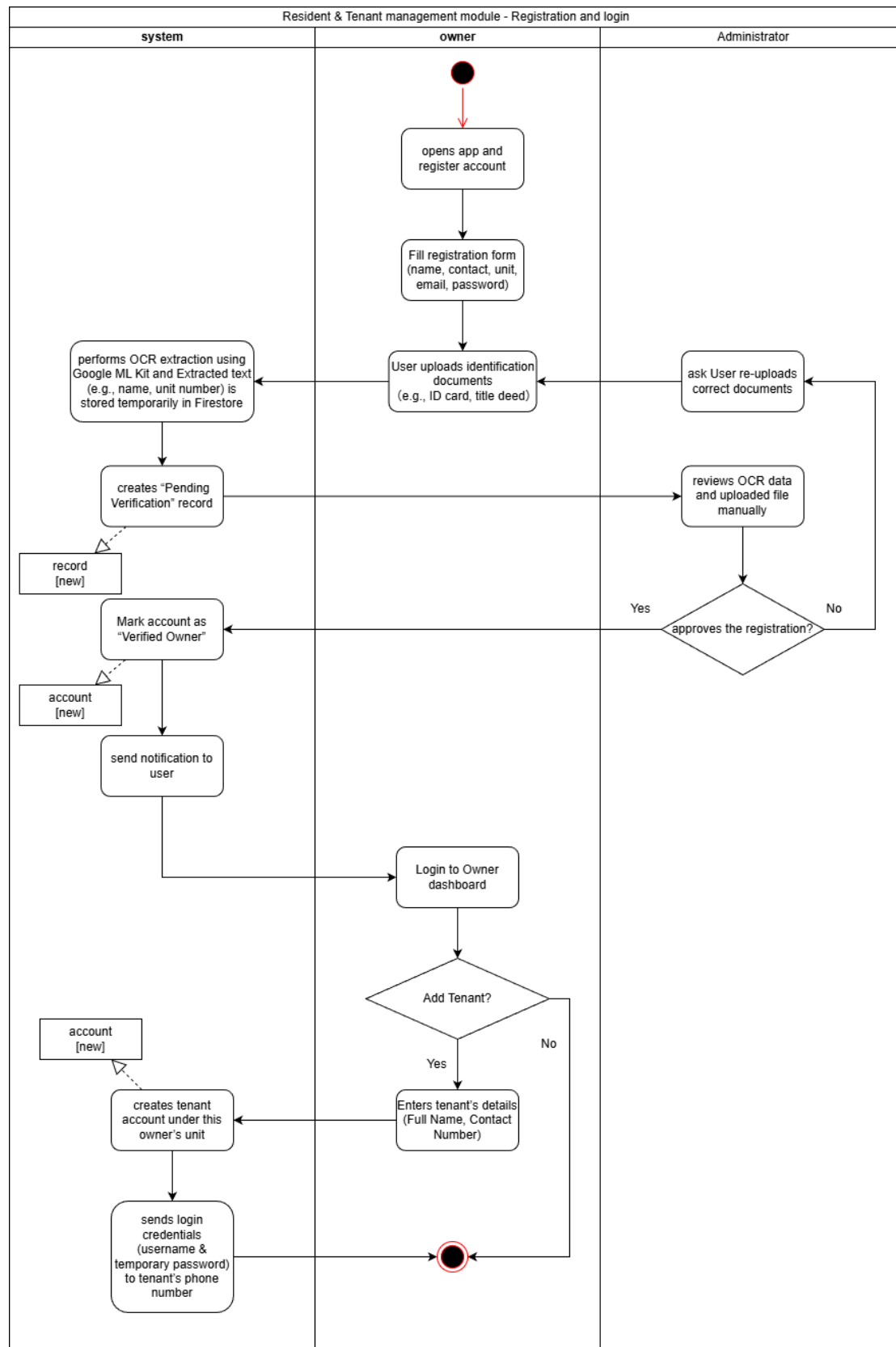
The diagram also highlights workflow relationships:

- Document verification and onboarding connect residents with ownershipDocument.
- Tenant lifecycle management is represented through the Resident–Tenant relationship.
- Visitor pre-registration and security screening are modeled through Visitors and EntryLog.
- Facility Booking incorporates conflict detection and approval workflows.
- The Admin module ties into announcements, feedback, and maintenance handling.
- PaymentTransaction supports blockchain-linked financial transparency.

Overall, the class diagram provides a complete blueprint of the system’s logical structure. It ensures that each module is modular yet interconnected, supporting maintainability, scalability, and clarity throughout the development process. This structural model forms the foundation for implementing system logic and ensures consistent alignment between functional requirements and application design

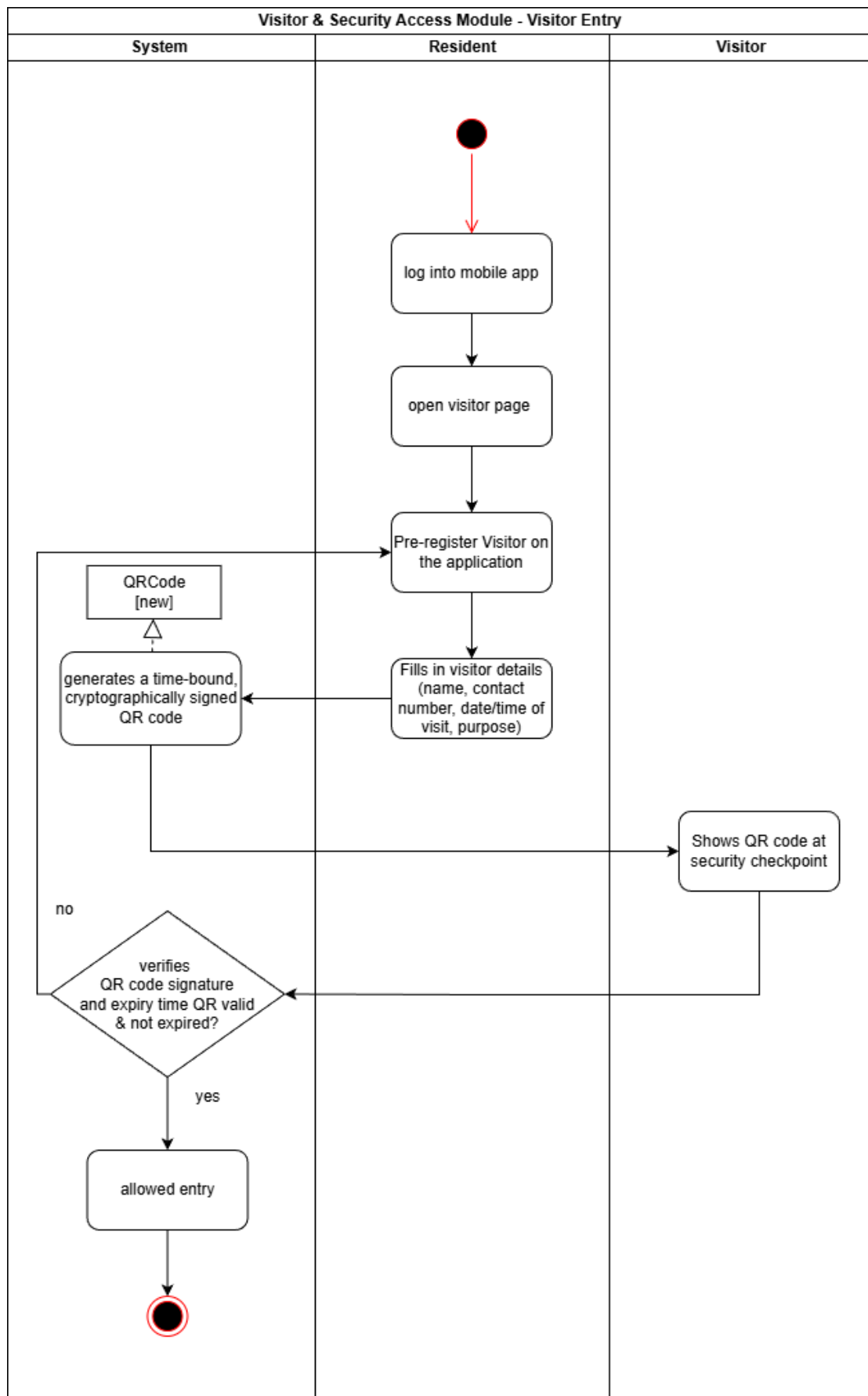
## 4.1.2 Activity diagram

## Resident &amp; Tenant management module - Registration and login



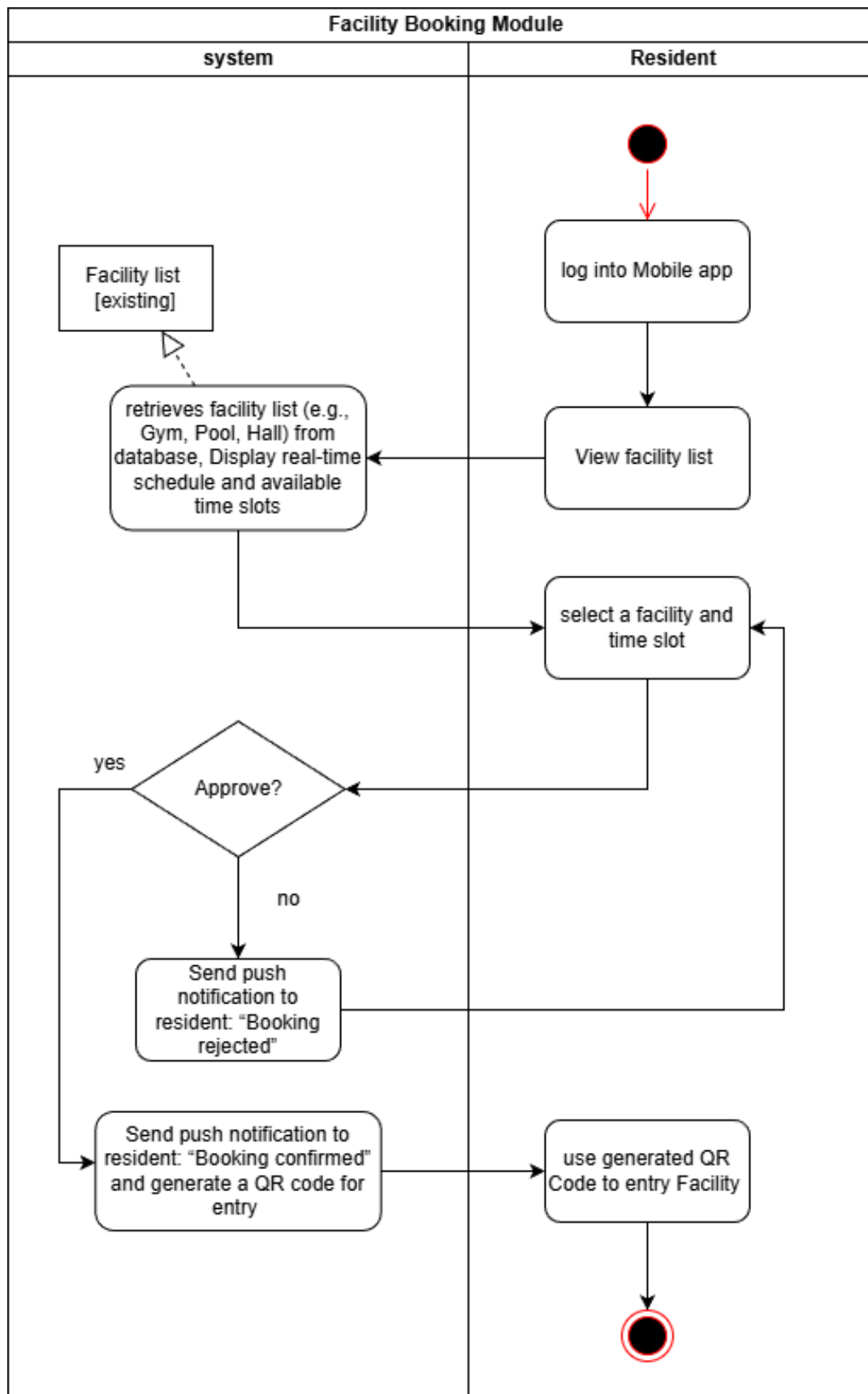
This flowchart illustrates the registration and login process for the Resident & Tenant Management Module, detailing the interaction between three main actors: the System, the Owner, and the Administrator. The process begins when an Owner registers by filling out a form and uploading identification documents. The System then performs OCR extraction using Google ML Kit to capture key details such as name and unit number, storing this data temporarily in Firestore. A "Pending Verification" record is created at this stage. The Administrator reviews the extracted OCR data and the uploaded documents. If discrepancies are found, the owner is asked to re-upload the correct documents. Once satisfied, the administrator approves the registration, marking the owner's account as "Verified," and a notification is sent to the user. Upon logging into the Owner Dashboard, the owner has the option to add a tenant. If selected, the owner enters the tenant's details (full name and contact number). The system then creates a tenant account under the owner's unit and sends login credentials, including a username and temporary password, to the tenant's phone number. This workflow ensures a structured and secure process for verifying property owners and onboarding tenants within the management system.

## Visitor &amp; Security Access Module - Visitor Entry



The Visitor Entry workflow illustrates the end-to-end process by which a resident pre-registers a guest and enables secure entry through a time-bound QR code. The process begins when the resident logs into the mobile application and navigates to the visitor management page. From there, the resident initiates a pre-registration request and fills in the required visitor details, including name, contact number, purpose of visit, and the intended date and time of arrival. Once submitted, the system automatically generates a cryptographically signed QR code that is uniquely bound to the provided visit information and configured with a strict validity timeframe.

When the visitor arrives at the condominium, they present the QR code at the security checkpoint. The security device or application scans the QR code and forwards it to the system for verification. The system then validates the QR code signature to ensure that it has not been tampered with and checks whether the code is still within its active validity period. If the verification fails—either due to expiry, an altered code, or an unregistered visit—the entry request is rejected. If the QR code is successfully validated, the system grants entry approval, allowing the visitor to proceed. This workflow enhances overall security by ensuring that all visitor entries are pre-authorized, traceable, and protected .

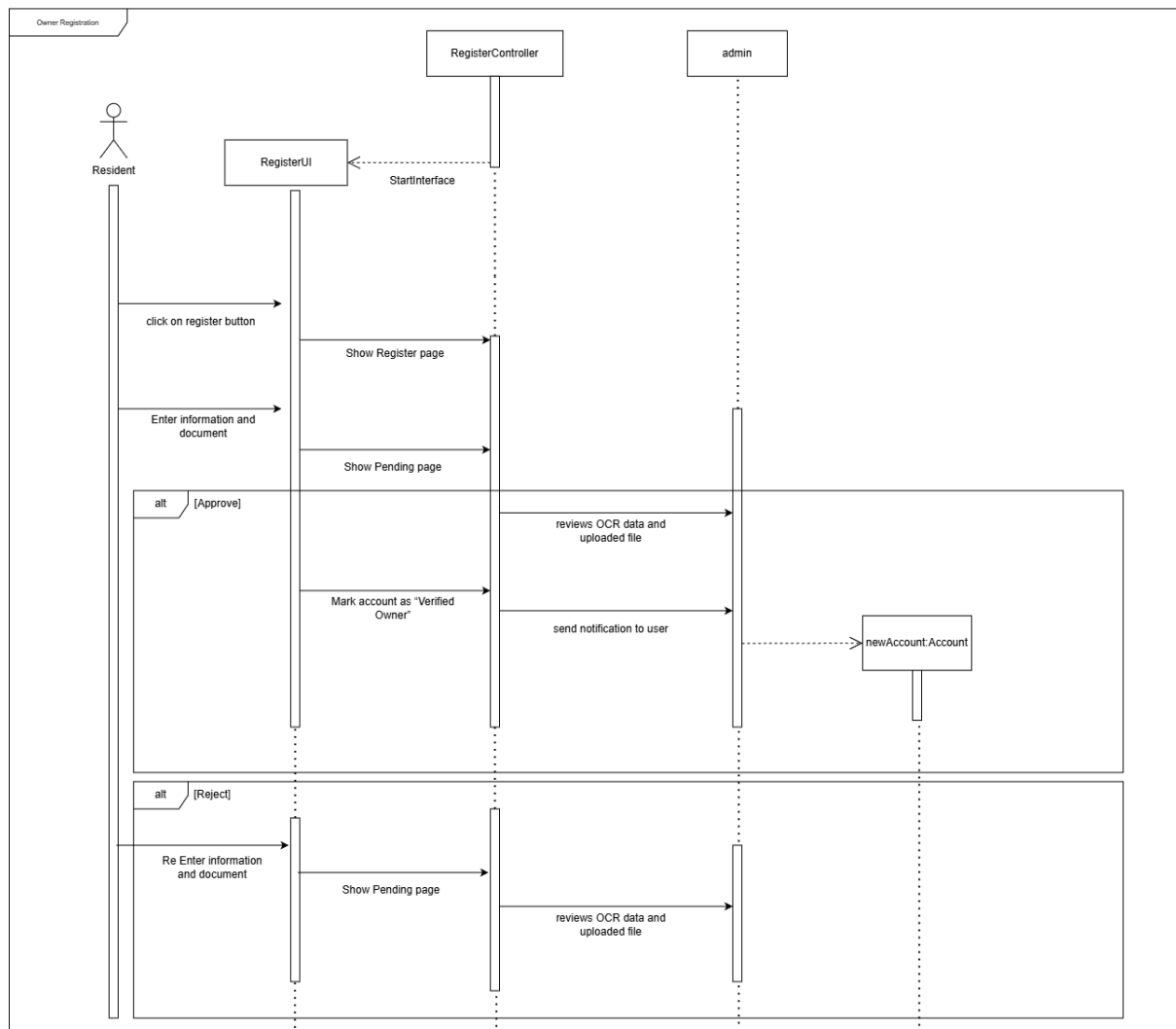
**Facility Booking Module - Booking Facility**

The Facility Booking Module flowchart illustrates the complete process through which a resident interacts with the system to make a facility reservation. The sequence begins when the resident logs into the mobile application and navigates to view the list of available facilities. Meanwhile, the system retrieves the facility information—such as gym, pool, or hall—from the database, including real-time availability and time slot schedules. After reviewing the options, the resident selects a specific facility and preferred time slot. The system then triggers an approval process, where the request is evaluated for conflicts, such as overlapping bookings or unavailable time slots. If the booking is rejected, the system sends an instant push notification informing the resident of the unsuccessful request. If the booking is approved, the system confirms the reservation and automatically generates a QR code for secure facility entry. A push notification containing the confirmation details and QR code is sent to the resident. The resident can subsequently use this QR code at the facility checkpoint to gain access. This workflow ensures an efficient, transparent, and secure end-to-end booking experience while preventing double-booking and maintaining real-time updates across the system.

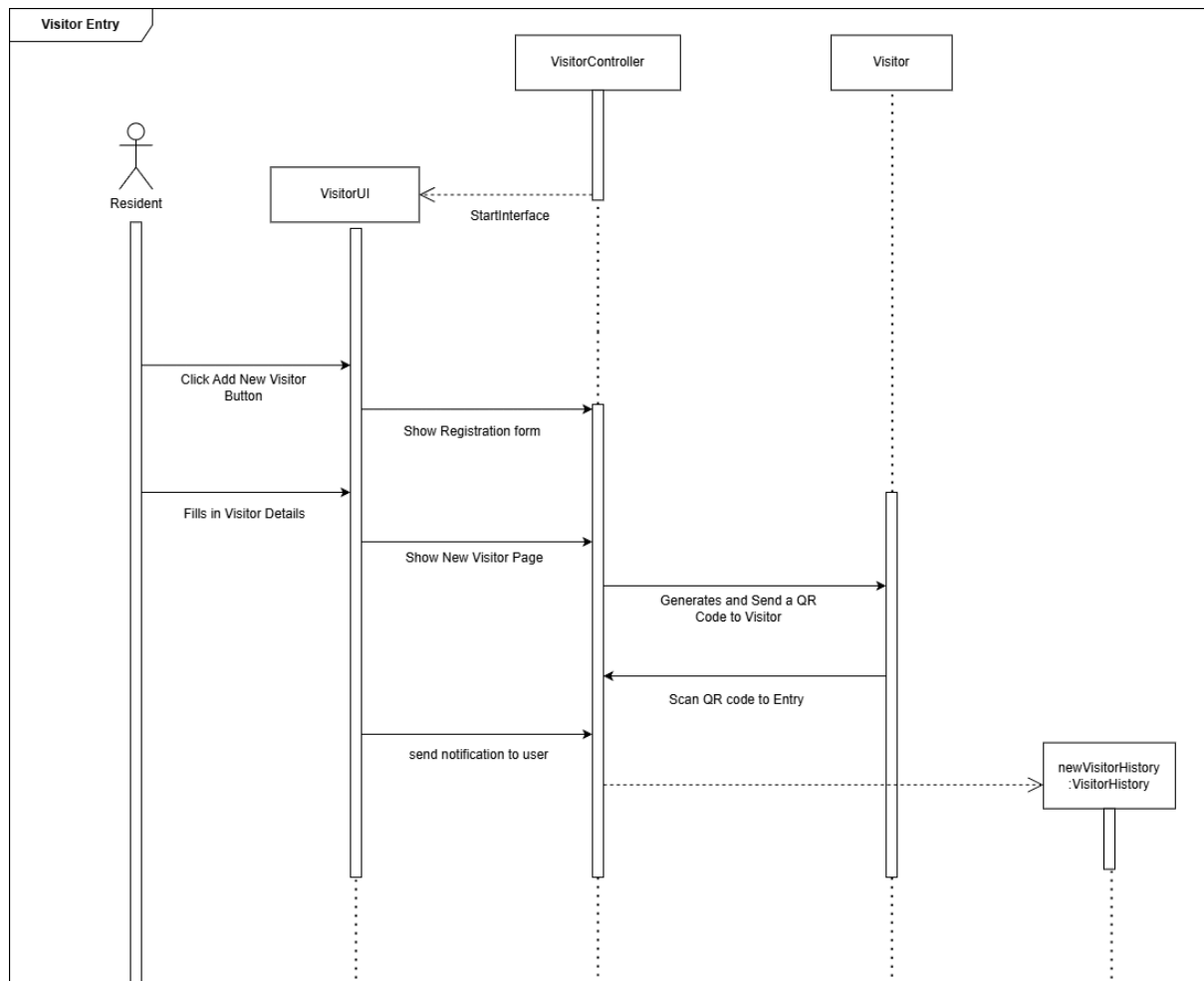
### 4.1.3 Sequence diagram

#### Resident & Tenant management module

#### Owner Registration sequence diagram



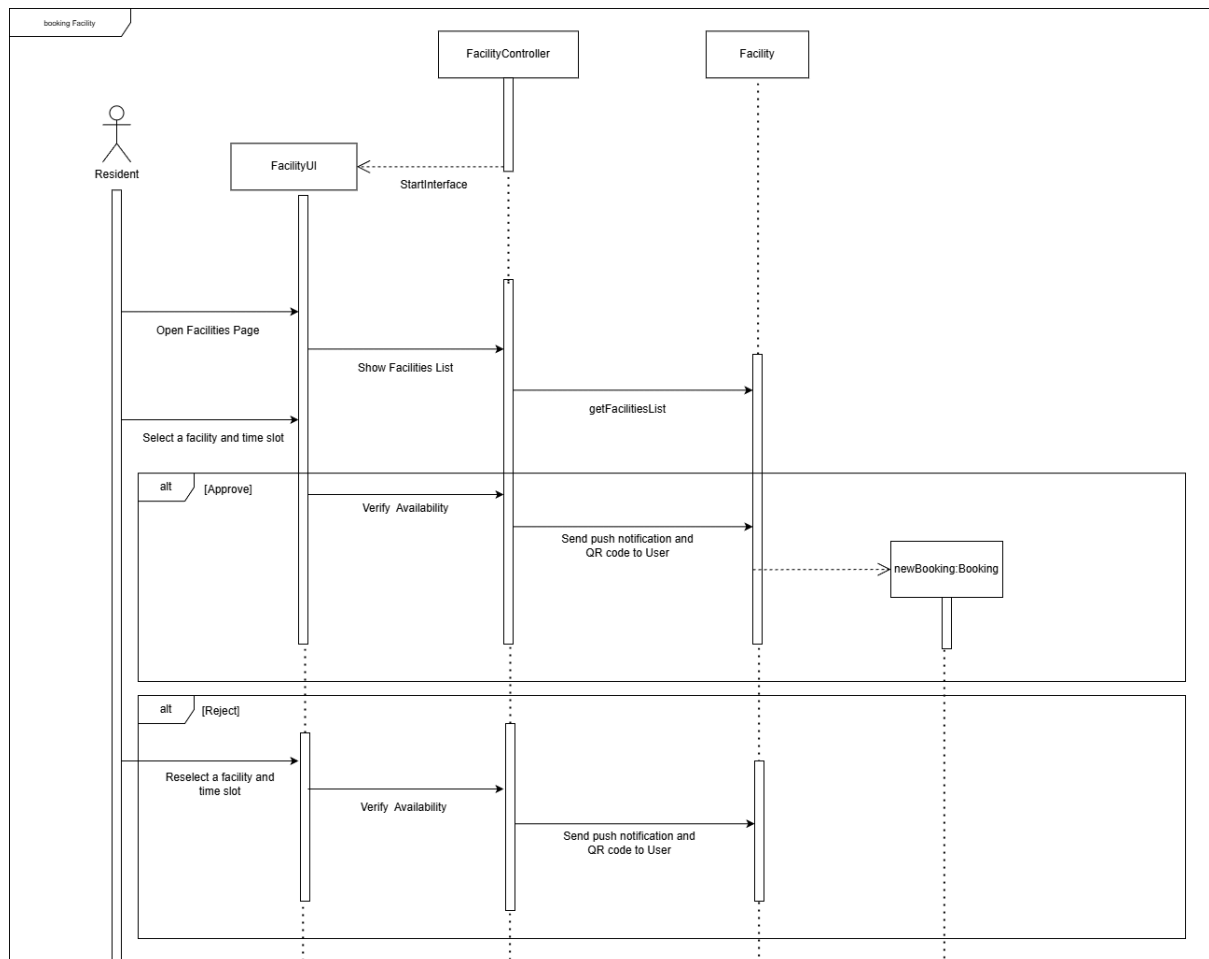
This sequence diagram illustrates the process of registering a visitor and generating a QR code for entry. The resident begins by accessing the visitor interface and submitting the visitor's details. The VisitorController processes the information and returns a new visitor page. The system then generates a secure QR code for the visitor, which is later scanned at the entry point. A notification is sent to the resident, and the visitor information is stored in the visitor history for security tracking.

**Visitor & Security Access Module****Visitor Entry Sequence Diagram**

This sequence diagram demonstrates the workflow for owner registration and admin verification. The resident starts by submitting personal details and uploading documents through the registration interface. The RegisterController forwards the request to the admin for OCR verification. The admin reviews extracted data and decides whether to approve or reject the registration. If approved, the system marks the user as a verified owner and notifies them; if rejected, the resident must re-enter the required information.

## Facility Booking Module

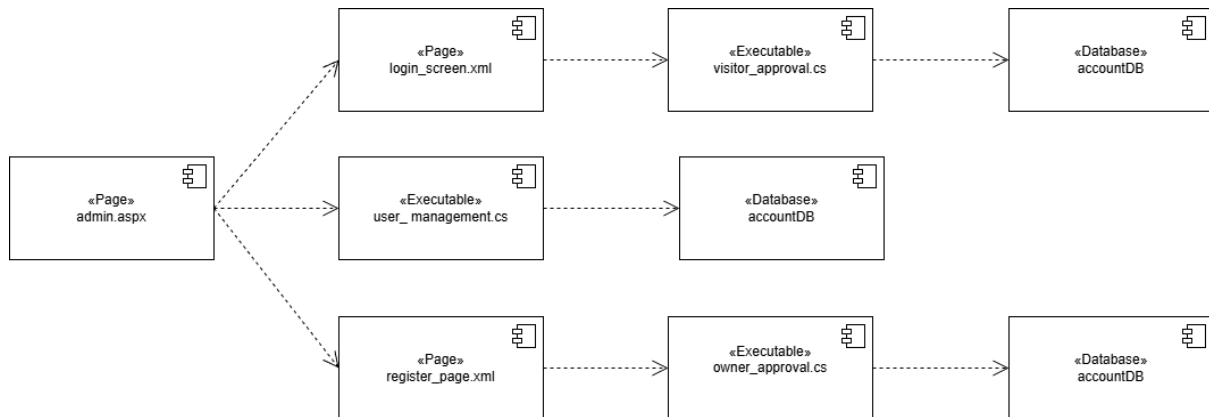
### Booking Facility



This sequence diagram presents the steps involved in booking a facility. The resident first opens the facilities page and selects a preferred facility and time slot. The FacilityController verifies availability before processing the booking. If the request is approved, a QR code and confirmation notification are sent to the resident, and the booking record is stored. If the booking is rejected, the resident can select another time slot and repeat the process.

#### 4.1.4 Component diagram

##### Resident & Tenant management module

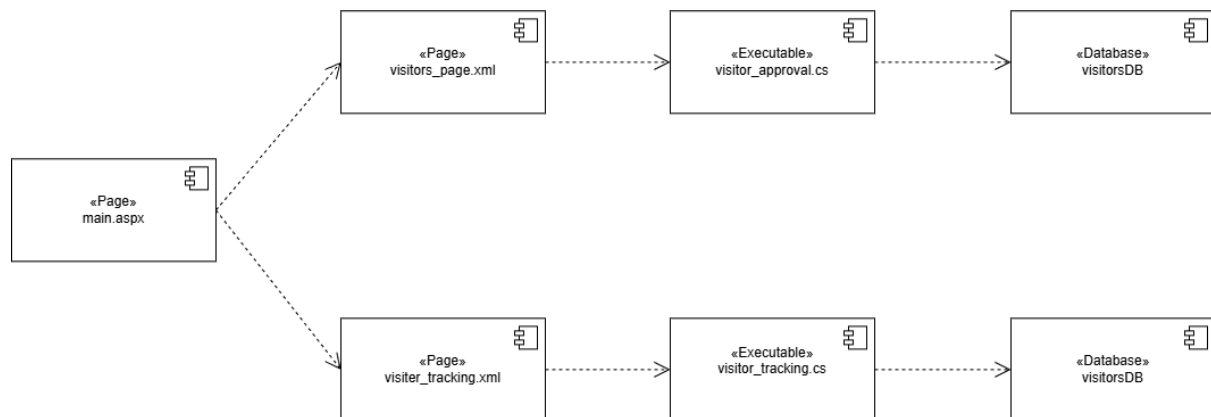


This component diagram illustrates the key software components involved in the Resident & Tenant Management Module and how they interact to support user onboarding, document verification, and tenant lifecycle management.

The Flutter Mobile App acts as the main interface for residents, allowing them to register, upload ownership documents, and manage tenant information. All data submitted through the mobile interface is processed using Firebase Authentication, which handles secure identity validation and user role assignment.

Uploaded documents such as title deeds and identification cards are stored in Firebase Cloud Storage, while extracted metadata and verification records are saved in Cloud Firestore. During onboarding, the system triggers Google ML Kit (OCR) or Google Cloud Vision API through Firebase Cloud Functions to automatically extract and validate text from documents. Administrators interact with the Admin Panel (also built on Flutter) to review the extracted data and approve or reject the onboarding request.

This module ensures secure onboarding, automated KYC processing, accurate document handling, and clear separation of responsibilities between residents and administrators.

**Visitor & Security Access Module**

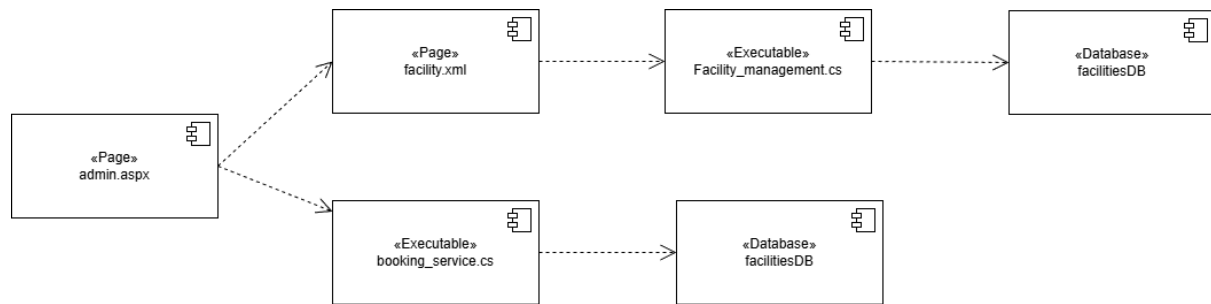
The component diagram for the Visitor & Security Access Module outlines the components involved in managing visitor registration, secure QR access, and real-time entrance verification.

Residents use the Flutter Mobile App to pre-register visitors and generate time-bound QR codes. The system stores visitor details and QR metadata in Cloud Firestore, while QR images are exported and delivered to users through the mobile interface. When a visitor arrives, the Security Scanner Device scans the QR code and communicates directly with the backend to verify validity.

Verification logic is executed by Firebase Cloud Functions, which validate cryptographic signatures, check expiry, and retrieve visitor information from Firestore. If approved, the result is shown instantly to the security personnel. The system then logs the entry event into Firestore for audit trail purposes.

This architecture ensures a secure, automated, and traceable visitor access workflow, replacing manual logbooks with digital verification backed by QR authentication and real-time cloud validation.

## Facility Booking Module



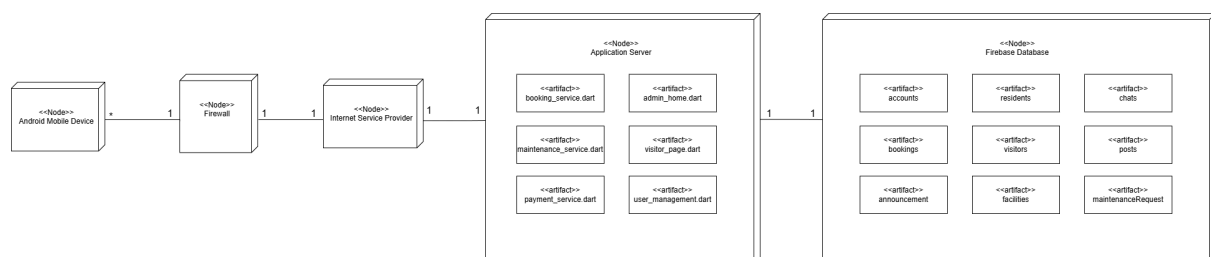
The component diagram for the Facility Booking Module presents the components required to support real-time booking, conflict detection, and automated facility approval workflows.

Residents initiate booking requests via the Flutter Mobile App, which retrieves live facility schedules and availability from Cloud Firestore. Once a user submits a request, the data is processed by Firebase Cloud Functions, which performs essential logic such as conflict checking, approval routing, and schedule updates.

Approved bookings are stored in Firestore, while associated QR codes for facility entry are kept in Cloud Storage. Residents receive booking updates through Firebase Cloud Messaging (FCM), ensuring instant confirmation or rejection notifications.

Administrators can modify schedules, approve bookings, or review usage logs through their version of the mobile interface. The combination of real-time Firestore updates and automated cloud functions ensures seamless, consistent, and error-free booking operations for all users.

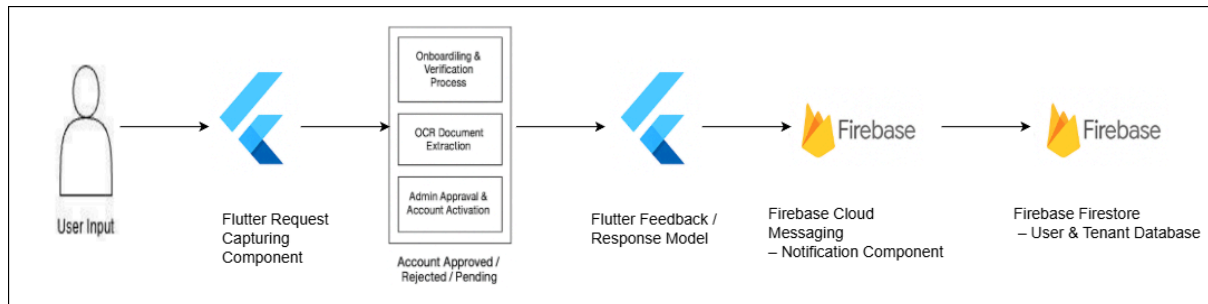
### 4.1.5 Deployment diagram



This deployment diagram illustrates how the Vanguard system operates across multiple nodes, beginning from the Android mobile device used by residents and security personnel. All requests are securely routed through the firewall and Internet Service Provider before reaching the application server, where core Dart services such as booking, visitor, and user management modules are executed. The system's data including accounts, residents, bookings, visitors, announcements, and maintenance requests, stored and managed within the Firebase Database, enabling real-time synchronization and reliable cloud-based operation.

## 4.2 System Architecture and Design

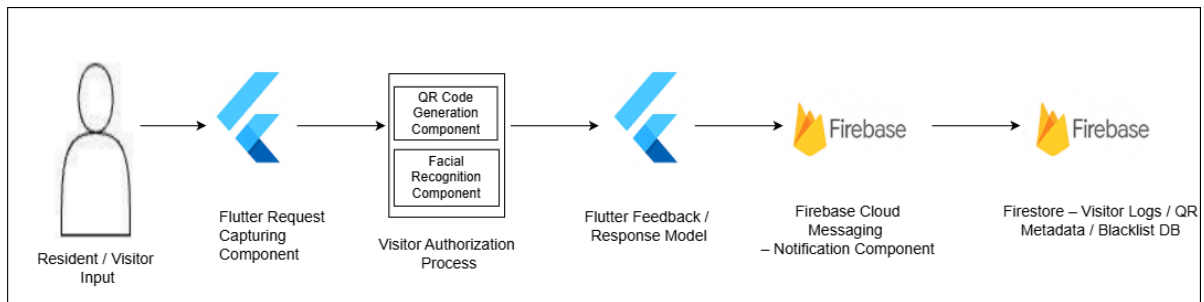
### Resident & Tenant Management Module



The Resident & Tenant Management Module forms a foundational component of the Vanguard system, responsible for handling user onboarding, identity verification, and the complete lifecycle management of tenants linked to each residential unit. This module directly addresses long-standing issues in condominium management such as manual verification of ownership documents, lack of standardized onboarding processes, and inaccurate resident records. Its core backend logic is structured as the grouped “Onboarding & Verification Process,” consisting of the OCR Document Extraction component and the Approval & Account Activation component. The OCR engine, powered by Google ML Kit, automatically extracts and validates textual information from title deeds, identification cards, and tenancy agreements, reducing human error and administrative processing time. The Approval & Account Activation component then enables administrators to review extracted data, cross-check unit ownership, and activate or reject accounts accordingly.

A separate feedback layer delivers status updates to residents throughout the onboarding process, providing clear indications of approval, rejection, or required resubmissions. This layered design ensures that interface-level changes do not interfere with backend processing logic, improving modularity and maintainability. The module interacts closely with Firebase Authentication for secure account creation and Firestore for storing validated ownership records, tenant profiles, and account states. By automating identity verification and structuring the tenant onboarding workflow, the module improves data accuracy, enhances security, and significantly reduces administrative workload while laying a scalable and reliable foundation for user identity management across the entire system.

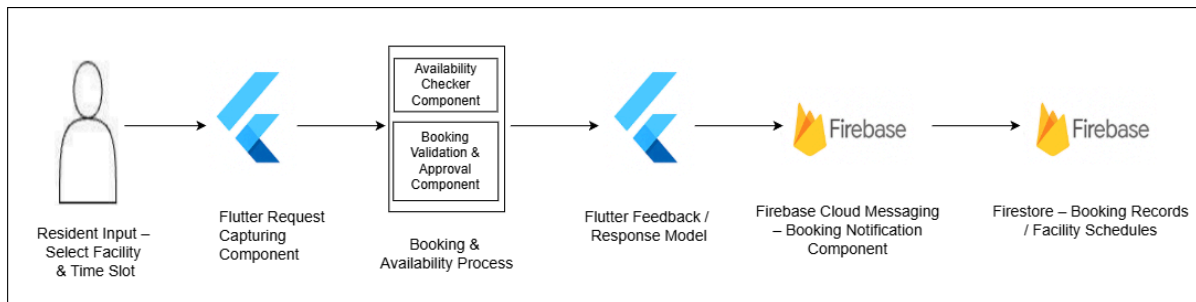
## Visitor & Security Access Module



The Visitor & Security Access Module provides a comprehensive and secure solution for managing visitor entry, replacing traditional manual logbooks with a fully digital and traceable access control system. The module solves key issues in existing condominium operations, such as unverified visitor access, incomplete visitor records, and inconsistent security procedures. Its backend is grouped under the “Visitor Authorization Process,” which consists of two main components: QR Code Generation and Facial Recognition. The QR Code Generation component creates cryptographically signed, time-bound QR passes for guests pre-registered by residents. These passes prevent unauthorized reuse, limit entry to specific time windows, and ensure traceability by linking each pass to a specific resident unit and visitor profile. Meanwhile, the Facial Recognition component utilizes ML Kit’s on-device processing to authenticate residents at access points, providing a touchless and secure entry alternative to keycards or PIN codes.

A dedicated feedback layer communicates registration confirmations, QR pass availability, and access outcomes to residents and visitors. Security personnel also receive real-time scanning results through the same layer, including visitor identity details, validation status, and access permissions. Tight integration with Firestore enables the storage of visitor logs, QR metadata, timestamps, and blacklist records, supporting centralized oversight and historical auditing. By combining automated authorization logic with structured data management, the module ensures a high level of security, enhances monitoring capabilities, and creates a reliable digital alternative to manual visitor handling methods.

## Facility Booking Module



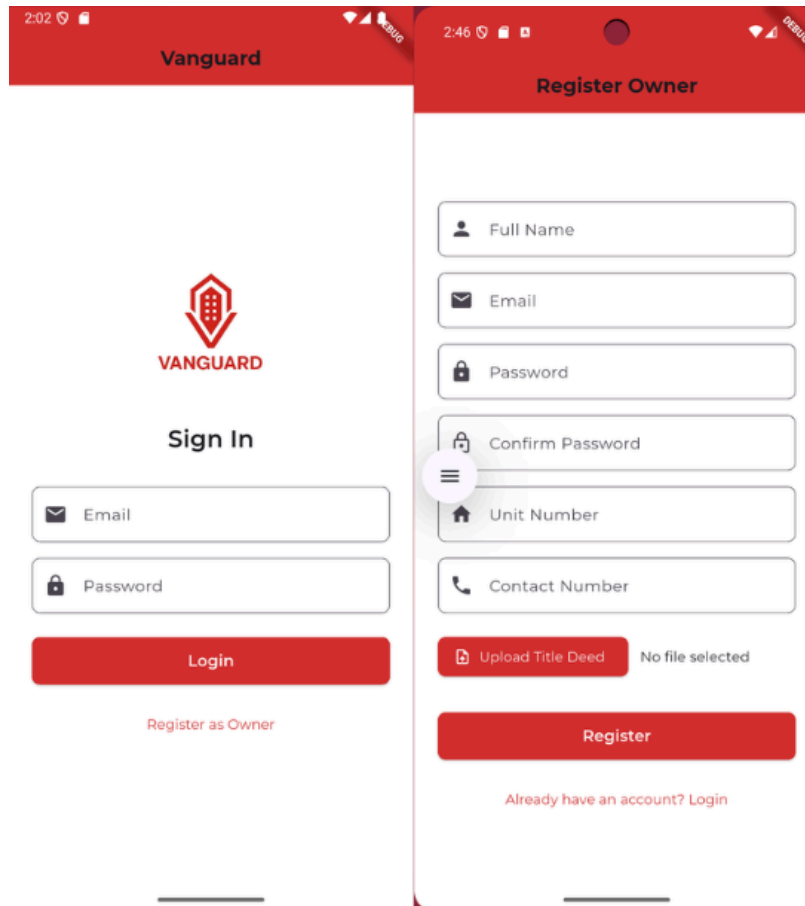
The Facility Booking Module manages the complete lifecycle of booking shared condominium facilities, addressing major operational challenges such as scheduling conflicts, double-bookings, and inefficient manual coordination. Its system architecture is based on the grouped “Booking & Availability Process,” which incorporates the Availability Checker and Booking Validation components. The Availability Checker retrieves real-time facility schedules from Firestore, ensuring that residents always have up-to-date information on open time slots. The Booking Validation component evaluates booking requests against existing reservations, enforces conflict-prevention rules, and handles approval workflows for facilities requiring administrative authorization. This logical grouping ensures that booking decisions are processed consistently and automatically, reducing the administrative burden traditionally placed on management staff.

A dedicated feedback layer informs residents of booking approvals, rejection reasons, and updates through push notifications and in-app status indicators. Administrators can review pending bookings and modify facility availability without affecting backend validation logic, ensuring clear separation between data presentation and processing. All booking data—including timestamps, facility metadata, conflict logs, and related QR facility-entry codes—is stored in Firestore to maintain consistency and integrity. This architecture supports efficient scheduling, eliminates common manual booking errors, and provides a scalable framework capable of accommodating additional facilities or expanded approval rules in the future.

## 4.3 User Interface

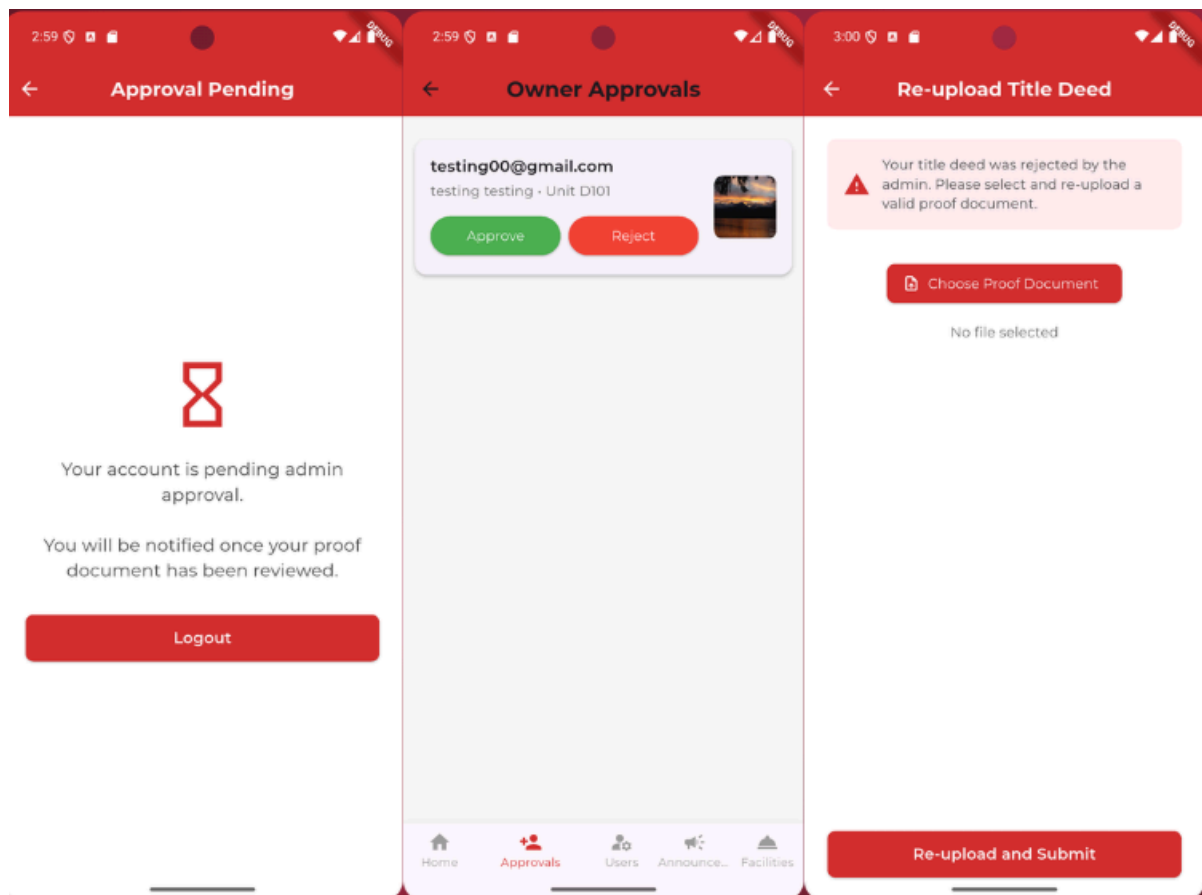
### Resident & Tenant Management Module

#### Login & Registration Page



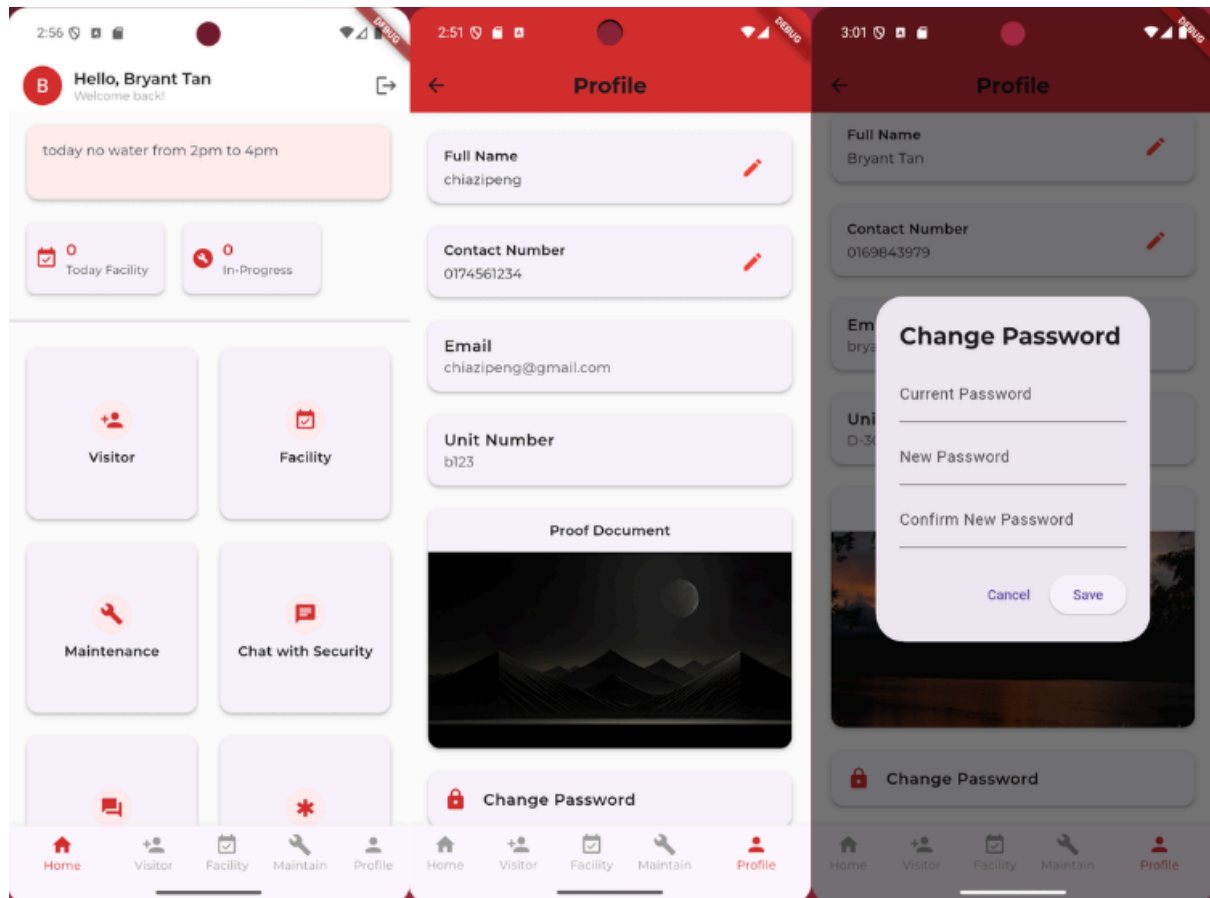
The Sign In and Register Owner screens provide the entry point for users of the Vanguard system. The Sign In screen allows existing users to log in using their email and password through a clean and simple interface. New property owners can proceed to the Register Owner screen, where they enter their personal details, unit information, and upload their title deed for verification.

### Owner Approval Pending Workflow

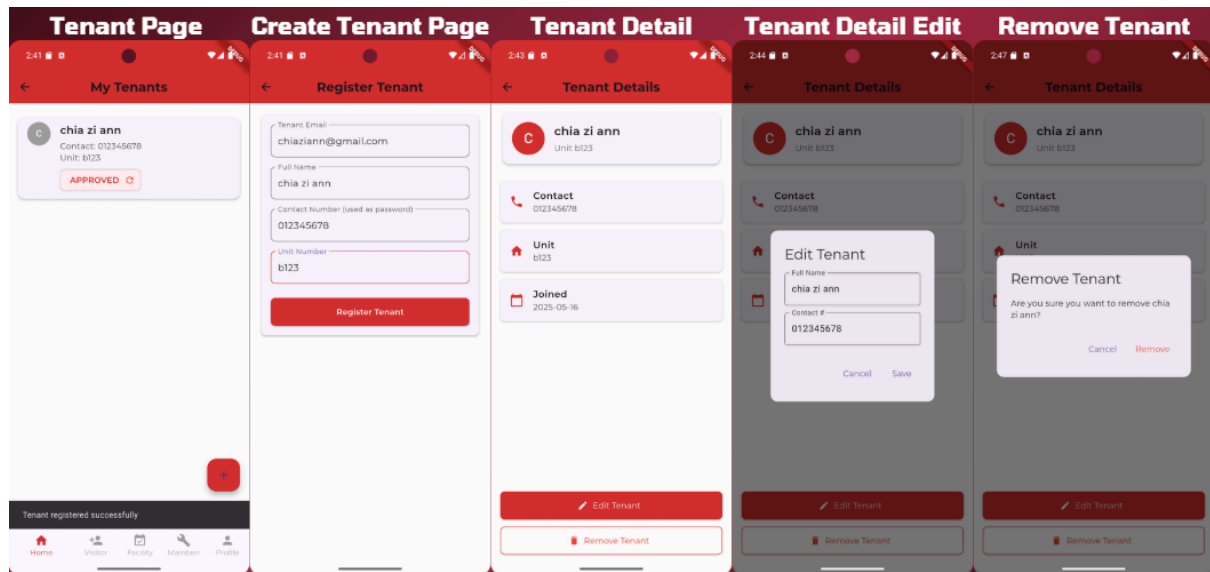


These interfaces support the owner verification workflow within the Vanguard system. The Approval Pending screen informs newly registered owners that their account is awaiting administrator review and restricts access until verification is complete. The Owner Approvals screen, accessible to administrators, displays pending applications along with options to approve or reject each submission based on the uploaded proof documents. If a submission is rejected, the Re-upload Title Deed screen allows the owner to upload a corrected or valid document, ensuring a smooth and secure verification process.

## User Homescreen & Profile Page



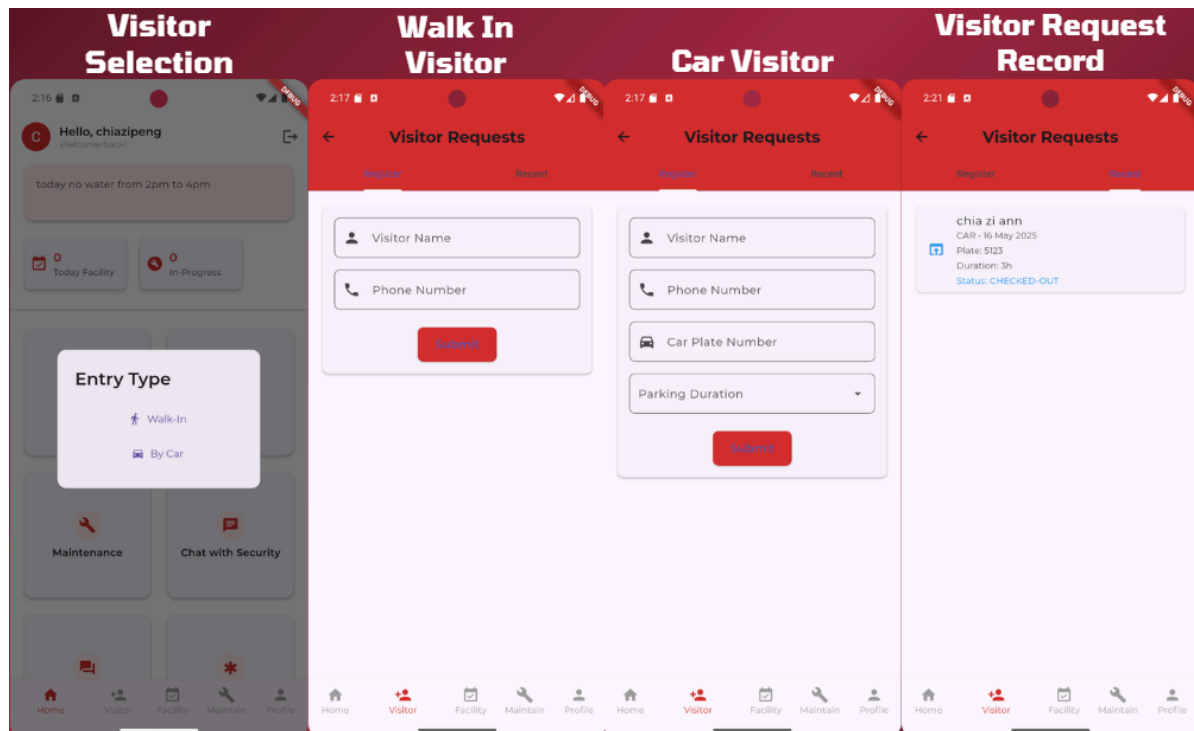
These screens represent the resident's dashboard and profile management features within the Vanguard system. The Home screen provides quick access to key modules such as visitor registration, facility booking, maintenance requests, and security chat, while also displaying announcements and task summaries. The Profile screen allows residents to view and update their personal information, unit details, and proof documents. Additionally, the Change Password interface enables users to securely update their credentials through a simple and user-friendly dialog, ensuring privacy and account protection.

**Tenant Management (Owner Side)**

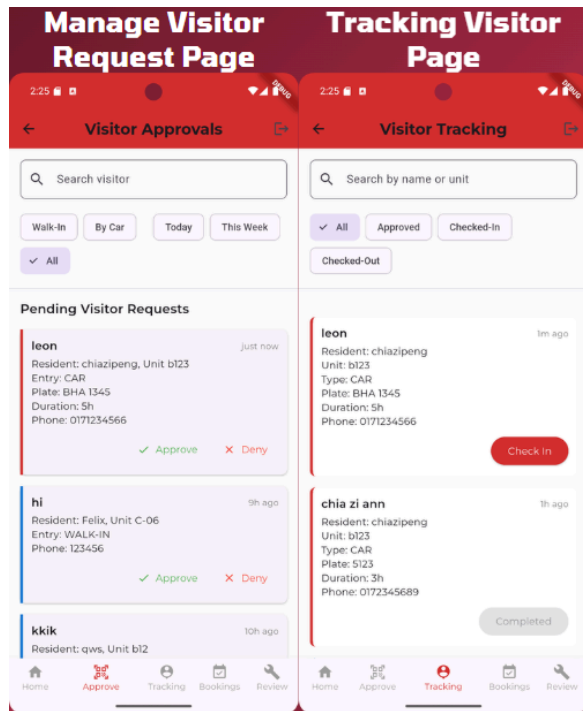
These screens represent the Tenant Management workflow available to verified property owners. The My Tenants page displays a list of registered tenants associated with the owner's unit. Through the Register Tenant page, owners can create a tenant account by providing basic details such as name, email, and contact number. The Tenant Details page allows owners to view tenant information, update their details through the Edit Tenant dialog, or remove a tenant when the tenancy ends. This interface supports a clear and efficient tenant lifecycle management process within the Vanguard system.

## Visitor & Security Access Module

### Visitor Registration (User Side)



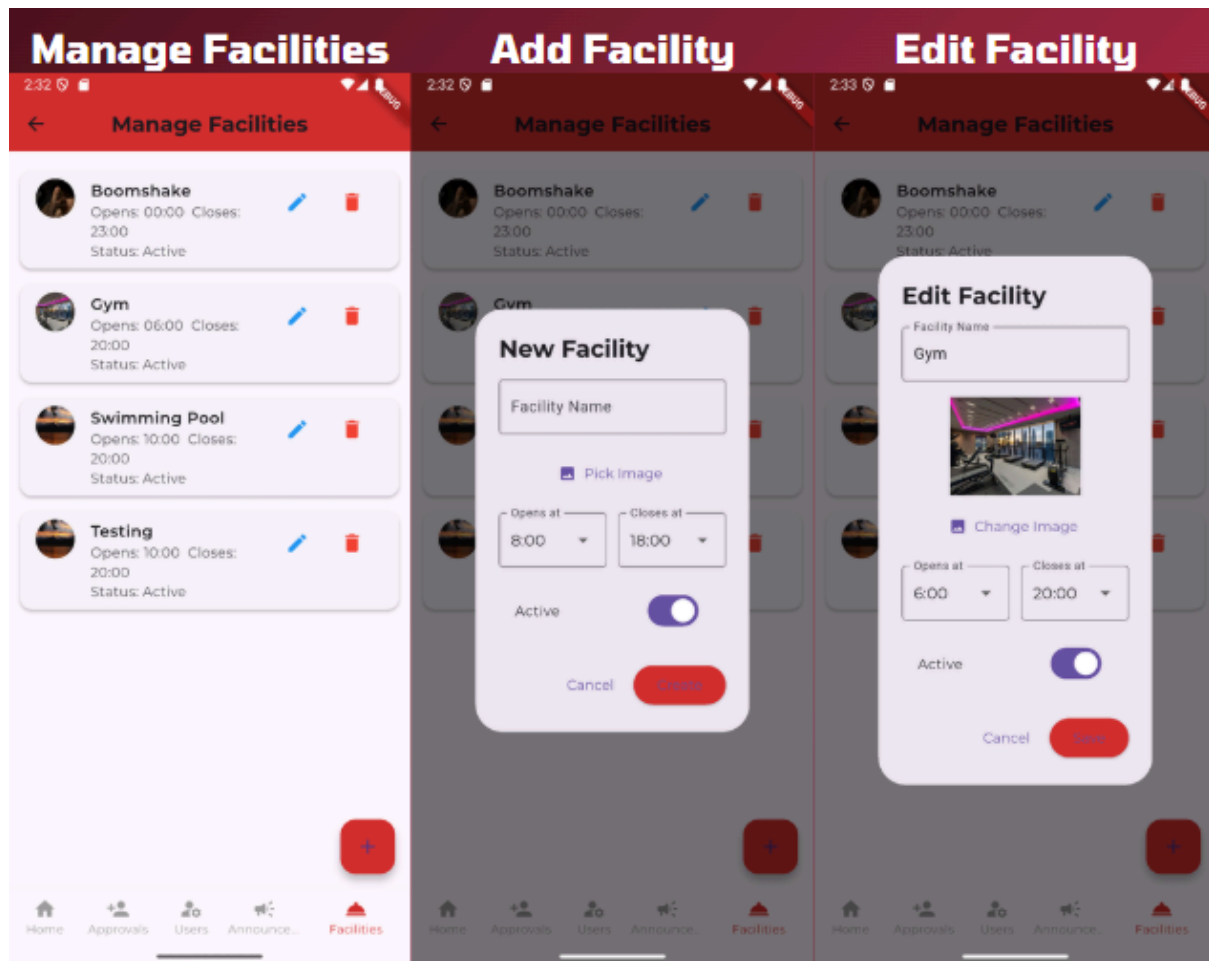
These screens illustrate the visitor registration workflow within the Vanguard system. Residents begin by selecting the visitor entry type—either walk-in or by car. For walk-in visitors, the system collects basic details such as name and phone number, while car visitors additionally provide a car plate number and parking duration. Once submitted, the request is recorded and displayed in the Recent tab, allowing residents to track visitor status in real time. This streamlined interface ensures quick and accurate visitor registration while improving transparency and security at the condominium entrance.

**Visitor Registration (Security Side)**

These screens support the security team's workflow for managing and monitoring visitor activity. The Visitor Approvals page allows security personnel to review pending visitor requests submitted by residents, with filters for entry type and time, and options to approve or deny each request. The Visitor Tracking page provides real-time monitoring of visitors currently checked in or previously checked out, allowing staff to verify details and update visitor status through the Check-In and Completed actions. Together, these interfaces enhance security oversight and ensure accurate, centralized visitor management.

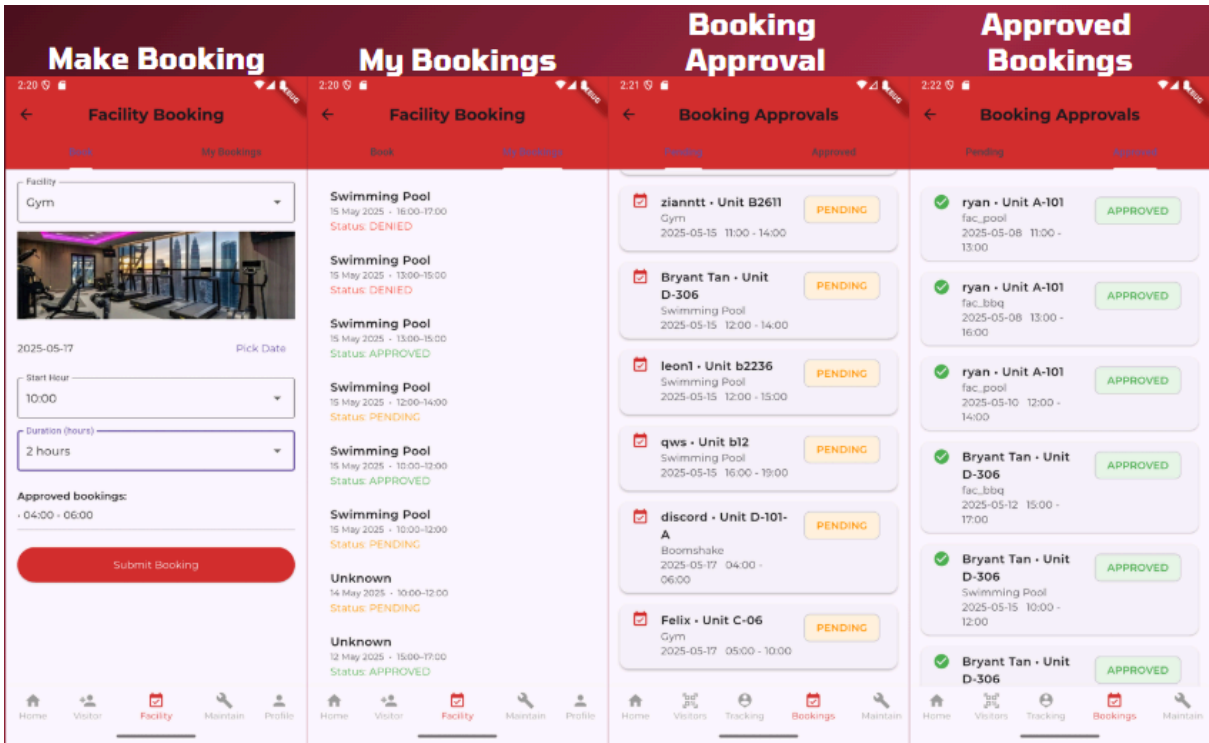
## Facility Booking Module

### Facility Booking (Admin Side)



These screens represent the facility management interface available to administrators within the Vanguard system. The Manage Facilities page displays all existing facilities along with options to edit or delete them. Administrators can create new facilities through the Add Facility screen by specifying the facility name, operating hours, status, and an image. The Edit Facility screen allows modifications to the facility's details, including updating images, adjusting opening and closing times, or changing its active status. This module enables efficient and flexible management of facility information to ensure accurate scheduling and smooth operations for residents.

Facility Booking



These screens illustrate the facility booking workflow within the Vanguard system. The Make Booking page allows residents to select a facility, choose a date and time slot, and submit a reservation request. The My Bookings page provides residents with a clear overview of all their past and upcoming bookings, including approval statuses. For administrators or security staff, the Booking Approval page displays pending booking requests that can be approved or rejected. Approved bookings are shown in a separate tab for easy tracking. Together, these interfaces provide a streamlined, transparent, and well-organized booking experience for both residents and management.

## 4.4 Chapter Summary and Evaluation

Chapter 4 presented a complete and detailed system design for the Vanguard Condominium Management System, establishing a clear and robust blueprint that connects the requirements identified in earlier chapters to their technical realization. Through a combination of UML diagrams, system architecture breakdowns, and user interface designs, the chapter demonstrated how each subsystem works cohesively to fulfill the goals of security, efficiency, scalability, and user-centered design defined at the outset of the project.

The UML diagrams provided multiple perspectives on the system's structure and behavior. The class diagram illustrated the logical organization of the system by modeling core entities such as Users, Tenants, Visitors, Bookings, Facilities, and EntryLogs, revealing the relationships and responsibilities that underpin key workflows. Activity diagrams captured the dynamic process flows within the Resident & Tenant Management, Visitor & Security Access, and Facility Booking modules, showing how decisions, conditions, and system interactions work together to support real-world condo operations. Sequence diagrams further decomposed these processes into clear message exchanges between user interfaces, controllers, backend services, and Firestore, making explicit how operations such as OCR verification, QR-based visitor entry, and facility booking approval unfold moment by moment.

Component diagrams reinforced the system's modularity by separating responsibilities across Flutter UI components, controllers, Firebase Authentication, Cloud Functions, Firestore, and ML Kit services. Each diagram highlighted the division between presentation, business logic, and data storage, demonstrating how Vanguard achieves maintainability and clean separation of concerns. The deployment diagram illustrates how the final system operates across mobile devices, cloud infrastructure, and security checkpoints, emphasizing real-time synchronization, secure communication channels, and reliance on Firebase's scalable cloud environment.

The User Interface designs translated these technical models into practical, intuitive screens that support residents, administrators, and security personnel in their daily tasks. Screens for onboarding, tenant management, visitor pre-registration, QR scanning, real-time visitor tracking, facility booking, and administrative facility control emphasized clarity, responsiveness, and accessibility. The consistent UI patterns across modules showcase the system's commitment to usability for users of all ages, including elderly residents, aligning well with the project's SDG-related objectives.

Overall, the system design strongly aligns with the functional and non-functional requirements established in Chapter 3. The modular architecture supports scalability, enabling additional features such as IoT integration or advanced analytics to be added in the future without disrupting core components. Automated processes—such as document OCR, facial recognition, conflict detection,

and push notifications—collectively reduce administrative workload, minimize human error, and enhance security. The reliance on Firebase for authentication, real-time data storage, and cloud functions ensures reliability, low latency, and secure handling of sensitive information.

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heduling%20conflicts%20happen%20when%20there,%2C%20academic%2C%20or%20social%20set  
tings.](https://oncehub.com/blog/common-causes-of-scheduling-conflicts-and-how-to-avoid-them#:~:text=Sc%20heduling%20conflicts%20happen%20when%20there,%2C%20academic%2C%20or%20social%20settings.)

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**Appendices****Appendices A : Form 1 (Student Details)****BMCS3403 PROJECT 1  
FORM 1 : STUDENT DETAILS**

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| 1. STUDENT DETAILS         |                                                                                    |                            |        |
|----------------------------|------------------------------------------------------------------------------------|----------------------------|--------|
| Student Name               | CHIA ZI PENG                                                                       |                            |        |
| Registration No            | 23WMR14662                                                                         |                            |        |
| Academic Year              | 2025/2026                                                                          | Cohort                     | 202309 |
| Programme                  | RSD - Bachelor of Information Technology (Honours) in Software Systems Development |                            |        |
| House Phone No (Permanent) | -                                                                                  | House Phone No (Temporary) | -      |
| Handphone No               | 017-9685185                                                                        |                            |        |
| Email (TAR UC)             | chiazp-wp22@student.tarc.edu.my                                                    |                            |        |
| Email (Personal)           | chiazipeng@gmail.com                                                               |                            |        |

**Appendices B : Form 2 (Project Proposal)**

|                        |                                                                                      |                        |                   |
|------------------------|--------------------------------------------------------------------------------------|------------------------|-------------------|
| <b>Student Name</b>    | <b>CHIA ZI PENG</b>                                                                  |                        |                   |
| <b>Programme</b>       | Bachelor of Information Technology (Honours) in Software Systems Development         |                        |                   |
| <b>Email</b>           | <a href="mailto:chiazp-wp22@student.tarc.edu.my">chiazp-wp22@student.tarc.edu.my</a> | <b>Mobile Phone No</b> | <b>0179685185</b> |
| <b>Supervisor Name</b> | <b>Ms. Jessie Teoh</b>                                                               |                        |                   |

**PROJECT TITLE**

<Condominium Management Application> : <Resident & Tenant Management, Visitor & Security Access Module, Facility Booking Module>

**ABSTRACT**

Traditional condominium management systems continue to face significant challenges including inefficient visitor tracking, manual facility booking processes, and fragmented resident data management. Vanguard presents a comprehensive mobile-based solution that revolutionizes property management through six integrated modules: (1) Resident & Tenant Management featuring secure onboarding with AI-powered document verification (KYC) and OCR technology, (2) Visitor & Security Access utilizing QR-based visitor passes coupled with facial recognition authentication, (3) Facility Booking with intelligent real-time scheduling and automated approval workflows, (4) Maintenance Module enabling seamless request submission and live status tracking, (5) Admin Dashboard offering advanced user management tools and data analytics, and (6) Payment System facilitating automated fee collection with complete transaction history.

Developed using Flutter for cross-platform compatibility and Firebase for robust backend services, Vanguard adopts a dual-application architecture: a Resident App and an Admin/Security App, each tailored to their specific user roles. The Resident App incorporates innovative communication features including in-app chat functionality with administrators and security personnel, as well as a feedback system for continuous service improvement. The Admin App includes comprehensive security management tools, allowing administrators to blacklist unauthorized visitors and monitor suspicious activities.

Key technological implementations include Firebase Authentication for secure identity management, Cloud Firestore for real-time data synchronization, Google Vision AI for biometric verification, and Firebase Cloud Messaging for instant notifications. The mobile-first approach ensures ubiquitous access, immediate responsiveness, and superior user experience compared to traditional web-based systems.

Anticipated outcomes demonstrate significant operational improvements: a 30-40% reduction in administrative workload through process automation, 25% faster incident resolution via real-time communication channels, and enhanced security through the integrated blacklisting system. By digitizing legacy processes while incorporating modern AI capabilities and mobile convenience, Vanguard establishes a new standard for smart property management that prioritizes both operational efficiency and resident satisfaction, all while maintaining strict compliance with data privacy regulations.

**PROBLEM**

Traditional condominium management systems suffer from inefficient processes, poor communication, and security vulnerabilities, leading to operational inefficiencies and resident dissatisfaction. Key issues include:

**-Manual Visitor Tracking**

Traditional log books are often riddled with inaccuracies. Handwriting clarity issues and missed entry fields compromise visitor data integrity (Xenia Team,2024).

Security personnel struggle to verify identities quickly, increasing risks of unauthorized access.

**-Disorganized Facility Booking**

Phone/email-based bookings result in scheduling conflicts and no-shows.(Harish Kannan R, 2025)

Staff waste significant time managing reservations manually(Bookingplanner,2022).

**-Fragmented Communication**

Residents lack direct channels to report issues or provide feedback, delaying response times(Timetec, 2025).

Maintenance requests are often lost in emails or phone calls.

**-Security Gaps**

No centralized system to blacklist suspicious visitors or track recurring offenders(Timetec, 2025).

Traditional keycards or PINs can be easily shared or stolen.

These problems highlight the need for an integrated, mobile-based solution that automates processes, enhances security, and improves resident engagement.

**SOLUTION****1. Resident & Tenant Management Module**

Validate proof: Owners upload proof (e.g., title deeds) via app; OCR (Optical Character Recognition) validates documents.

Tenant Lifecycle Control: Lease-linked accounts with auto-expiry upon lease termination.

**Critical Functions:**

Document upload (Firebase Storage).

Profile CRUD operations with role-based access (Firebase Auth).

**2. Visitor & Security Access Module**

QR-Based Visitor Passes: Time-bound JWT tokens prevent pass reuse.

Face Recognition: Residents gain touchless entry using ML Kit's on-device face detection.

**Critical Functions:**

Real-time entry/exit logs (Firestore geotagged timestamps).

Blacklist alerts for banned visitors (Cloud Functions triggers).

**3. Facility Booking Module**

Dynamic Scheduling: AI-driven conflict detection (e.g., double-bookings).

**Critical Functions:**

Calendar integration with push notifications (Firebase Cloud Messaging).  
Admin dashboard for overriding bookings (Firebase Admin SDK).

#### 4. Maintenance Module

Residents can submit and track requests.  
Admins assign tasks and update statuses in real time.

#### 5. Admin Dashboard

Centralized user management (view/edit residents, staff).  
Analytics for tracking facility usage, visitor logs, and payments.

#### 6. Payment System

Automated fee collection with transaction history.

### **TARGET MARKET**

The primary target market for this condominium management system includes residents, property management teams, and security personnel, all of whom will benefit from its streamlined digital solutions.

Residents and tenants (both owners and renters) form the core user base, as they require convenient access to facilities (e.g., gyms, pools, and event spaces) and a secure way to manage visitors. The system eliminates the need for physical passes or manual booking processes, offering instant digital verification and real-time availability updates.

Property management teams (including administrators and maintenance staff) will use the platform to automate tedious tasks such as visitor log maintenance, booking approvals, and resident verification. By reducing manual workload, the system minimizes human errors and operational delays, improving overall efficiency.

Security personnel (guards and front-desk staff) will rely on the app for quick identity checks via QR codes or facial recognition, ensuring only authorized individuals enter the premises. The digital audit trail also simplifies incident reporting and compliance tracking.

Secondary stakeholders include property developers and joint management bodies (JMB), who can integrate this system into future smart condo projects, and service providers (e.g., cleaners and technicians), who can schedule visits seamlessly through the platform.

### **COMPETITION / CONTRIBUTION**

Compared to existing solutions like JagaApp which primarily focus on visitor management and basic facility booking, our condominium management system offers a more comprehensive and integrated approach. While competitors provide fragmented tools for individual tasks, our platform combines resident management, visitor access control, facility booking, maintenance coordination, payment processing, and advanced analytics into a single seamless solution. This end-to-end integration eliminates the need for multiple disconnected systems while improving efficiency through automation and real-time data synchronization.

The system's key contributions include: (1) a 30-40% estimated reduction in administrative workload through automated workflows, (2) enhanced security with biometric verification and digital audit trails, and (3) improved resident experience through mobile-first convenience. By

addressing both operational and user experience pain points that partial solutions overlook, our platform sets a new standard for modern, intelligent condominium management.

## MILESTONES

**Proposed Development Model / Research Method :** Agile methodology will be used, allowing iterative development and weekly review. Development will follow a modular approach with continuous integration and testing

### Project Schedule:

| ACTIVITIES                                             | EXPECTED OUTCOME                                | COMPLETION DATE |
|--------------------------------------------------------|-------------------------------------------------|-----------------|
| Prepare and Submit Proposal (Form 1–3)                 | Official proposal submission                    | 25/07 /2025     |
| Submit Chapter 1 – Introduction                        | Background, objectives & scope of modules       | 22/08/2025      |
| Submit Chapter 2 – Research Background                 | Literature review & competitor analysis         | 08/09/2025      |
| Submit Chapter 3 – Methodology & Requirements Analysis | Detailed development plan regarding the project | 22/09/2025      |
| Submit Chapter 4 – System Design                       | Wireframes, mockups, architecture plan          | 21/11/2025      |
| Submit Project I Portfolio                             | Complete documentation (Chapter 1 - 4)          | 12/12/2025      |
| Prepare Test Plan / Preview                            | Verified test scenario & user flows             | 20/02/2026      |
| Final System Testing                                   | Fully functional system                         | 06/03/2026      |
| Submit Draft FYP Report                                | Chapters 1-7 with implementation details        | 03/04/2026      |
| Submit Final Report & Source Code                      | Complete FYP documentation and ready to submit  | 13/04/2026      |

## REFERENCES

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**Appendices C : Form 3 (Proposal Moderation)**

| 1. PROJECT DETAILS                       |                                                                                                                                    |                       |              |
|------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------|-----------------------|--------------|
| <b>Student Name</b>                      | CHIA ZI PENG                                                                                                                       | <b>Student ID</b>     | 23WMR14662   |
| <b>Programme</b><br>(Circled The Option) | RST / REI / <b>RSD</b> / RIS / RSW / RDS / RIT                                                                                     | <b>Cohort</b>         | 202309       |
| <b>Supervisor Name</b>                   | Ms. Jessie Teoh                                                                                                                    | <b>Moderator Name</b> | Ms. Mawardah |
| <b>Project Title / Individual Title</b>  | Condominium Management System                                                                                                      |                       |              |
| <b>Category</b>                          | 6.1 <b>Development</b><br><br>6.2 <b>Research</b><br><br><i>Client Name (if the project is a collaboration with the industry):</i> |                       |              |

| ITEM                                                                                                                                                                           | DESCRIPTOR                                                                                                                                                                                                                                                                                                                                                                                |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Problem and Scope Formulation                                                                                                                                                  | The problem identified in this project :<br>5. <i>reflects a real-world problem or is supported by literature</i><br>6. <i>is unambiguous and has a clearly defined scope that is suitable for FYP</i><br>7. <i>can be associated with one or more of the United Nations' Sustainable Development Goals (SDG)</i><br>8. <i>has the complexity and scale which are appropriate for FYP</i> |
| Feasibility and Practicality                                                                                                                                                   | The proposed solution :<br><ul style="list-style-type: none"> <li><i>can be developed within the FYP time frame</i></li> <li><i>consists of modules which can be divided between the team members (for group work)</i></li> <li><i>is relevant to the identified problem</i></li> <li><i>is suitable to be implemented effectively in real-world scenarios</i></li> </ul>                 |
| Creativity and Innovation                                                                                                                                                      | The project :<br>7.1 <i>includes new or unique features which distinguish the project from other similar solutions</i><br>7.2 <i>offers afresh perspective or approach to solving the identified problem</i><br>7.3 <i>leveraging IT technologies and the latest trends in software development</i>                                                                                       |
| Contribution and Reach                                                                                                                                                         | The proposed solution :<br><ul style="list-style-type: none"> <li><i>will be easily adopted by the targeted users</i></li> <li><i>will contribute towards the betterment of the community or the targeted users</i></li> <li><i>has high potential for commercialisation or to be presented in technology based showcases</i></li> </ul>                                                  |
| <b>Note:</b><br>7- very satisfied      6 - satisfied      5 - slightly satisfied      4- neutral<br>3 - slightly dissatisfied      2 - dissatisfied      1 - very dissatisfied |                                                                                                                                                                                                                                                                                                                                                                                           |

### 3. FEEDBACK

| COMMENTS AND CHANGES RECOMMENDED BY MODERATOR | ACTIONS TAKEN (SUPERVISOR) |
|-----------------------------------------------|----------------------------|
|                                               |                            |

**MODERATED BY**

Moderator Signature :

Date :

## RECEIVED BY \_\_\_\_\_

Supervisor Signature :

Date :

**Appendices E : Form 4 : Project Appointment Record**

| Student's Details |                                                                                    |
|-------------------|------------------------------------------------------------------------------------|
| Name              | Chia Zi Peng                                                                       |
| Programme         | RSD - Bachelor of Information Technology (Honours) in Software Systems Development |

| No | Date   | Time    |         | Duration<br>(Minutes) | Venue  | Project Progress & Remarks                             | Students'<br>Signature |
|----|--------|---------|---------|-----------------------|--------|--------------------------------------------------------|------------------------|
|    |        | From    | To      |                       |        |                                                        |                        |
| 1. | 30July | 8:30pm  | 9:02pm  | 32                    | Online | Go through proposal and make correction                | <i>CHIA</i>            |
| 2. | 6Aug   | 2:40pm  | 3:23pm  | 43                    | Office | Final check of proposal and briefing for Chapter 1 & 2 | <i>CHIA</i>            |
| 3. | 20Aug  | 12:00pm | 12:33pm | 33                    | Office | Briefing for Chapter 1-4                               | <i>CHIA</i>            |
| 4. | 3Sep   | 3:00pm  | 3:39pm  | 39                    | Online | Checking for Chapter 1                                 | <i>CHIA</i>            |
| 5. | 14Nov  | 10:00am | 10:45am | 45                    | Office | Checking for Chapter 2 & 3                             | <i>CHIA</i>            |
| 6. | 4Dec   | 12:00pm | 12:20pm | 20                    | Office | Checking for Chapter 4                                 | <i>CHIA</i>            |
| 7. | 5Dec   | 12:00pm | 12:30pm | 30                    | Online | Perform prototype to supervisor                        | <i>CHIA</i>            |
|    |        |         |         |                       |        |                                                        |                        |
|    |        |         |         |                       |        |                                                        |                        |
|    |        |         |         |                       |        |                                                        |                        |
|    |        |         |         |                       |        |                                                        |                        |
|    |        |         |         |                       |        |                                                        |                        |

**Appendices B : Originality Report**

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12/12/2025, 23:33

FYP Portfolio 1 (CHIA ZI PENG)

## Originality report

---

**COURSE NAME**

202505 Final Year Project

**STUDENT NAME**

ZI PENG CHIA

**FILE NAME**

FYP Portfolio 1 (CHIA ZI PENG)

**REPORT CREATED**Dec 12, 2025

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### Summary

|                       |    |      |
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| oncehub.com            | 1 | 0.1% |

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